

AERODYNAMIC PERFORMANCE AND OPERATING-POINT OPTIMISATION

INVERTED NACA 6412 AEROFOIL

CFD • PYTHON POST-PROCESSING • XFOIL NUMERICAL VERIFICATION



Pressure field at the 16° maximum-downforce operating point.

DOCUMENT	VALUE
Author	Toby Mason
Document type	Engineering portfolio technical report
Study condition	50 m/s $Re = 3.424 \times 10^6$ $M = 0.147$
Issue	Final portfolio report
Date	24 September 2026

ENGINEERING POSITION

This report presents a two-dimensional performance-aerodynamics investigation inspired by Formula 1 wing-section requirements. It does not claim that the NACA 6412 is an actual Formula 1 profile or disclose proprietary motorsport geometry.

Document Control

ITEM	DETAIL
Title	Aerodynamic Performance and Operating-Point Optimisation of an Inverted NACA 6412 Aerofoil
Subtitle	A CFD, Python Post-Processing and XFOIL Numerical Verification Study at 50 m/s
Prepared by	Toby Mason
Status	Portfolio issue
Confidentiality	Personal engineering portfolio
Primary tools	Microsoft Excel, CATIA V5, Simcenter STAR-CCM+, Python and XFOIL 6.99

Revision Record

REV	DATE	DESCRIPTION
A	24 Sep 2026	Initial final portfolio issue assembled from archived simulation, post-processing and verification evidence.

Statement of Technical Integrity

Results are reported from the supplied simulation outputs and post-processing files. Unknown solver details have not been invented. The exact $k-\epsilon$ variant and wall-treatment option were not retained in the available evidence and are therefore declared as unverified. XFOIL is used as an independent numerical comparison, not as experimental validation. The finite-span, vehicle-interaction and transient-flow effects relevant to a complete Formula 1 aerodynamic device are outside the scope of this two-dimensional section study.

Evidence Base

The report integrates the supplied coordinate dataset, CATIA geometry image, STAR-CCM+ mesh and convergence screenshots, corrected coefficient CSV files, Python post-processing code and outputs, XFOIL run log, polar file and verification metrics.

Executive Summary

A traceable simulation and data-analysis workflow was developed to evaluate an inverted NACA 6412 aerofoil as a Formula 1-inspired downforce section. The profile was selected because its strongly cambered shape provides a useful accessible analogue for studying high-loading motorsport aerofoil behaviour. This is a design motivation rather than a claim of direct equivalence to a contemporary Formula 1 wing profile.

A 1 m chord coordinate-driven model was created in CATIA V5 and analysed in Simcenter STAR-CCM+ at 50 m/s, corresponding to Reynolds number 3.424×10^6 and Mach number 0.147. A 50 m × 50 m far-field domain was used with the aerofoil rotated about its leading-edge region. Steady two-dimensional RANS simulations employed a k-ε turbulence model. Each design was run for 1,000 iterations. [3]

The angle-of-attack sweep covered 0°, 5°, 10°, 15°, 16°, 17°, 18°, 19° and 20°. Coefficients were normalised to a consistent 1.000 m² reference area using preserved dimensional forces. Python automated data checks, coefficient conversion, force calculation, plotting, run metadata and comparison against XFOIL.

HEADLINE DECISION

15° is recommended as the balanced high-downforce operating point. It delivers 99.97% of the computed maximum downforce achieved at 16°, while reducing drag coefficient by 10.47% relative to 16° and retaining greater margin before the post-peak loading decline.

DECISION OBJECTIVE	AOA	KEY RESULT
Maximum aerodynamic efficiency	10°	CL /CD = 34.93
Balanced high-downforce operation	15°	CL = -1.7859; CD = 0.0635
Absolute maximum downforce	16°	CL = -1.7863; 2,735.3 N/m

The CFD and XFOIL datasets agreed on a peak-loading angle of 16°. Loading RMSE was 0.1879 and the linear loading-gradient difference was -7.09%. Drag agreement was weaker, with CFD consistently predicting higher drag. Strict lift mesh independence was not demonstrated: the medium-to-fine lift difference was approximately 10.4%, although drag differed by only approximately 1.55%. The selected medium mesh was retained because it produced the strongest steady-state convergence and enabled a computationally efficient design sweep. These limitations are carried into the recommendation.

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Nomenclature

SYMBOL	DEFINITION	UNIT
α	Angle of attack	deg
c	Aerofoil chord	m
C_D	Drag coefficient	–
C_L	Lift coefficient; negative values denote downforce	–
D	Drag force per metre span	N/m
L	Lift force per metre span	N/m
M	Mach number	–
q^∞	Freestream dynamic pressure	Pa
Re	Reynolds number	–

SYMBOL	DEFINITION	UNIT
Sref	Reference area	m ²
ρ	Air density	kg/m ³
μ	Dynamic viscosity	Pa·s

1. Introduction

1.1 Engineering Context

Aerodynamic downforce is a primary performance lever in single-seat motorsport because increased tyre normal load can raise the available cornering, braking and traction capability. The local performance of a wing section is governed by its geometry, angle of attack, Reynolds number, boundary-layer state and interaction with surrounding components. In a complete Formula 1 car, these effects are highly three-dimensional and tightly coupled. A two-dimensional section study remains valuable as a controlled environment for developing the core engineering workflow used to quantify loading, drag, separation and operating-point trade-offs.

The NACA 6412 was selected because its pronounced camber and 12% thickness distribution provide an accessible profile with qualitative similarity to the high-loading curved sections commonly associated with motorsport wing concepts. Public coordinate data identify the profile as having 6% maximum camber at approximately 40% chord and 12% maximum thickness near 30% chord. The section was inverted so that the generated lift acted as downforce. The selection provides a relevant learning platform, but the profile is not represented as a current or proprietary Formula 1 aerofoil. [1]

1.2 Research Question

RESEARCH QUESTION

How does angle of attack influence the downforce, drag and aerodynamic efficiency of an inverted NACA 6412 aerofoil at 50 m/s, and which investigated operating point provides the strongest engineering balance between aerodynamic loading, drag and separation margin?

1.3 Aim

The aim was to develop and apply a traceable computational workflow for evaluating an inverted NACA 6412 aerofoil across a controlled angle-of-attack range, combining coordinate-driven CAD, steady-state CFD, mesh sensitivity, automated Python post-processing and independent XFOIL comparison to support an evidence-based operating-point recommendation.

1.4 Objectives

1. Generate a traceable 1 m chord aerofoil from coordinate data using Excel and CATIA V5.
2. Develop a consistent STAR-CCM+ model at 50 m/s, $Re = 3.424 \times 10^6$ and $M = 0.147$.
3. Assess coarse, medium and fine mesh behaviour and select a working discretisation.
4. Investigate 0° to 20° incidence, including a refined sweep around the peak-loading region.
5. Normalise lift and drag outputs to a common 1.000 m^2 reference area.
6. Automate quality checks, calculations, plots and verification metrics in Python.
7. Compare the CFD trends with an XFOIL 6.99 viscous polar.
8. Define operating points for efficiency, balanced loading and maximum downforce.

1.5 Scope

INCLUDED	EXCLUDED
Inverted NACA 6412, 1 m chord	Proprietary Formula 1 geometry
2D steady RANS section analysis	Finite-span and induced-drag effects
0° to 20° angle-of-attack sweep	Ground effect and car-body interaction
Mesh sensitivity and convergence evidence	Wind-tunnel or track validation
Python data pipeline and XFOIL comparison	Structural deformation and aeroelasticity
Forces per metre span	Transient URANS, DES or LES

2. Aerodynamic Basis

2.1 NACA 6412 Definition

In four-digit NACA notation, the first digit defines maximum camber as a percentage of chord, the second identifies the camber position in tenths of chord, and the final two digits define maximum thickness as a percentage of chord. NACA 6412 therefore represents 6% maximum camber at 40% chord with 12% maximum thickness. Inverting the profile reverses the sign of the generated lift under the global force convention, producing downforce. [1]

2.2 Governing Performance Quantities

$$q_{\infty} = \frac{1}{2} \rho V_{\infty}^2 \quad (1)$$

For $\rho = 1.225 \text{ kg/m}^3$ and $V_{\infty} = 50 \text{ m/s}$, the dynamic pressure is 1,531.25 Pa.

$$C_L = L / (q_{\infty} S_{\text{ref}}) \quad \text{and} \quad C_D = D / (q_{\infty} S_{\text{ref}}) \quad (2)$$

The common reporting area was set to 1.000 m^2 . For a two-dimensional section with 1 m chord, this is equivalent to reporting loads per metre span. Downforce coefficient is presented as $-C_L$ or $|C_L|$ where positive magnitude aids comparison.

$$Re = \rho V_{\infty} c / \mu \quad (3)$$

Using $c = 1 \text{ m}$ and $\mu = 1.789 \times 10^{-5} \text{ Pa}\cdot\text{s}$ gives $Re = 3.424 \times 10^6$.

$$\text{Aerodynamic efficiency} = |C_L| / C_D \quad (4)$$

Maximum efficiency, maximum loading and best race-car operating compromise are not necessarily the same point. A robust recommendation must therefore state its objective and consider sensitivity to separation.

2.3 Essential Mathematical Snapshot

The complete pre- and post-simulation calculation pack is supplied separately within the portfolio. The calculations below provide the most decision-critical mathematical checks directly within the main report.

CHECK	EQUATION AND SUBSTITUTION	RESULT
Reynolds number	$Re = \rho V c / \mu = (1.225 \times 50 \times 1.000) / (1.789 \times 10^{-5})$	3.424×10^6
Mach number	$M = V/a = 50/340.3$	0.147
Dynamic pressure	$q_{\infty} = \frac{1}{2} \rho V^2 = 0.5 \times 1.225 \times 50^2$	1,531.25 Pa
Common 2D reference area	$S_{\text{ref}} = c \times 1 \text{ m} = 1.000 \times 1.000$	1.000 m^2
16° corrected lift coefficient	$C_L = L / (q_{\infty} S_{\text{ref}}) = -2735.3 / (1531.25 \times 1)$	-1.7863
16° corrected drag coefficient	$C_D = D / (q_{\infty} S_{\text{ref}}) = 108.6 / (1531.25 \times 1)$	0.07091
16° aerodynamic efficiency	$ C_L / C_D = 1.7863 / 0.07091$	25.19

Pre-Simulation Benchmark

Thin-airfoil theory was used as an ideal attached-flow benchmark, not as an expected exact solution at 16°. Applying the NACA 6412 mean-camber-line method gives an upright-section zero-lift angle of approximately -6.232°. When the profile is inverted and evaluated at the simulated orientation, the ideal loading magnitude is:

$ CL_{ideal} = 2\pi \alpha - \alpha_L=0 $	$ CL_{ideal} = 2.438$
$ L'_{ideal} = CL_{ideal} q_\infty c$	$ L'_{ideal} = 3,732.9 \text{ N/m}$

The final CFD downforce at 16° was approximately 2,735.3 N/m, 26.72% below the inviscid thin-airfoil benchmark. This reduction is physically plausible because the analytical model excludes viscosity, finite thickness effects and high-incidence separation.

Reference-Area Correction Check

The raw STAR-CCM+ coefficient reports were internally consistent with their configured directional areas, but those areas were unsuitable for directly comparing lift and drag. Dimensional forces were therefore retained and both coefficients were recomputed using the same 1.000 m² area.

QUANTITY	RAW REPORT BASIS	COMMON-AREA RESULT
Lift / downforce	CL = -1.85672 using Ay = 0.9618438 m ²	CL = -1.7863 using Sref = 1.000 m ²
Drag	CD = 0.219149 using Ax = 0.3235639 m ²	CD = 0.07091 using Sref = 1.000 m ²
Physical forces	L = -2735.3 N/m; D = 108.6 N/m	Preserved unchanged

	MATHEMATICAL ROLE These calculations provide scale, sign and coefficient-consistency checks. The separate mathematical report retains the complete derivation, assumed force envelope, circulation calculation, pressure-coefficient relation and pre-/post-simulation comparison.
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2.4 Stall and High-Incidence Behaviour

As incidence increases, the pressure differential and downforce initially rise. The adverse pressure gradient also strengthens, making boundary-layer separation more likely. Once separation growth offsets the loading benefit, downforce reaches a maximum and declines while drag increases. In steady RANS, highly separated flow may appear as non-convergent or oscillatory force histories, indicating that a transient method may be more appropriate.

3. Engineering Workflow

The project was structured as a closed-loop engineering workflow rather than a sequence of isolated software tasks. Geometry traceability, numerical credibility, automated post-processing and independent comparison were maintained through each stage.

STAGE	TOOL	OUTPUT / CONTROL
1. Coordinate definition	Excel / CSV	62 planar coordinates in millimetres
2. Geometry generation	CATIA V5	Closed 1 m chord aerofoil and source CAD
3. CFD preparation	STAR-CCM+	Domain, boundaries, mesh and force reports
4. Design sweep	STAR-CCM+	Nine 1,000-iteration operating points
5. Data correction	Python	Common 1 m ² coefficient basis
6. Analysis	Python	Plots, forces, efficiency, ranked points
7. Verification	XFOIL 6.99	Independent viscous polar at matched Re and M
8. Decision	Engineering review	Efficiency, balanced and peak-loading recommendations

PORTFOLIO RELEVANCE

The workflow demonstrates the motorsport engineering disciplines of configuration control, repeatable simulation, objective-led optimisation, data quality assurance and transparent treatment of model limitations.

3.1 Tool Selection and Justification

TOOL	WHY SELECTED	ENGINEERING VALUE
Microsoft Excel / CSV	Transparent preparation, inspection and transfer of coordinate data.	Creates a controlled geometry source and supports repeatable macro input.
CATIA V5	Parametric CAD and Generative Shape Design capability.	Converts coordinate data into a closed 3D part and neutral exchange geometry.
Simcenter STAR-CCM+	Integrated meshing, RANS solution, design studies, reports and scenes.	Provides a consistent CFD workflow from meshing through design comparison.
Python	Flexible and auditable engineering-data environment.	Automates QA, correction, calculations, plots, metrics and traceability records.
XFOIL 6.99	Established low-order viscous aerofoil-analysis method.	Provides an independent numerical trend comparison at matched Re and M.

3.2 Implemented Automation: Excel to CATIA V5

The geometry stage used a macro-assisted coordinate-driven CAD method. NACA 6412 data were obtained in CSV form and organised into consistent X, Y and Z columns. The Excel/CATIA macro read the coordinate rows and created the corresponding points in CATIA V5 Generative Shape Design, avoiding the need to place or transcribe the points manually. The imported point set was then used to form the aerofoil boundary, close the profile and generate the 1 m chord three-dimensional source part. The completed geometry was retained as a CATIA part and exported as STEP for use in STAR-CCM+.

MACRO WORKFLOW STEP	AUTOMATED OR CONTROLLED ACTION	BENEFIT
1. Prepare source data	Arrange ordered X, Y and Z coordinates in a consistent worksheet / CSV structure.	Creates a traceable single source of geometry data.
2. Read coordinate rows	Macro iterates through the full coordinate list.	Reduces transcription risk and operator time.
3. Create CATIA points	Coordinates are instantiated in Generative Shape Design on the required plane.	Provides repeatable construction from the same dataset.
4. Build the profile	Imported points support spline creation, inspection and section closure.	Preserves the defined shape and enables geometry checks.
5. Create and export part	Closed 1 m chord section is formed into a 3D part and exported as CATPart and STEP.	Produces native and neutral files for downstream CFD.

This method is important for aerodynamic design studies because geometry updates can be introduced at the source-data level and reproduced without rebuilding every point manually. It improves speed, consistency and accuracy, while creating a clear link between the coordinate dataset and the final CAD model. The implemented work demonstrates macro-assisted import and geometry creation; it does not overclaim a live software link or a fully automated optimisation loop.

3.3 Automation Position Across the Project

WORKFLOW AREA	POSITION IN THIS PROJECT	FUTURE INDUSTRIAL EXTENSION
Excel to CATIA	Implemented macro-assisted coordinate import and repeatable geometry creation.	Parameterise aerofoil family, chord, camber and thickness; batch-build variants.
STAR-CCM+ CFD	Design study, simulations and exports completed through the available workflow.	Use Java macros to create cases, update AoA, mesh, run, check convergence and export results.
Python post-processing	Implemented end-to-end for data QA, coefficient correction, metrics and figures.	Connect to an automated case database and generate controlled report packs.
Engineering interpretation	Completed using engineering judgement and evidence review.	Retain human technical approval even when draft reporting is automated.

3.4 STAR-CCM+ Automation Opportunity

A future STAR-CCM+ Java macro could convert the present design-study process into a repeatable batch workflow. The macro could read an angle-of-attack schedule, rotate the aerofoil or update the relevant parameter, regenerate the mesh, initialise and run each case, evaluate residual and force-monitor acceptance criteria, export reports and monitor histories, write pressure and velocity scenes with fixed camera positions and colour scales, and save each case under a controlled naming convention. Failed or oscillatory cases should be flagged for review rather than silently accepted.

PROPOSED JAVA-MACRO FUNCTION	CONTROL PURPOSE
Case generation and AoA update	Removes repetitive setup and ensures identical treatment between designs.
Automated meshing and quality checks	Applies consistent controls and records cell-count and quality metrics.
Run control and convergence logic	Uses residual and force-monitor criteria rather than iteration count alone.
Report extraction	Exports forces, coefficients and monitor histories consistently.
Scene export	Writes pressure, velocity, mesh and convergence images with controlled views.
Logging and exception handling	Records failures, warnings, runtime and configuration for traceability.

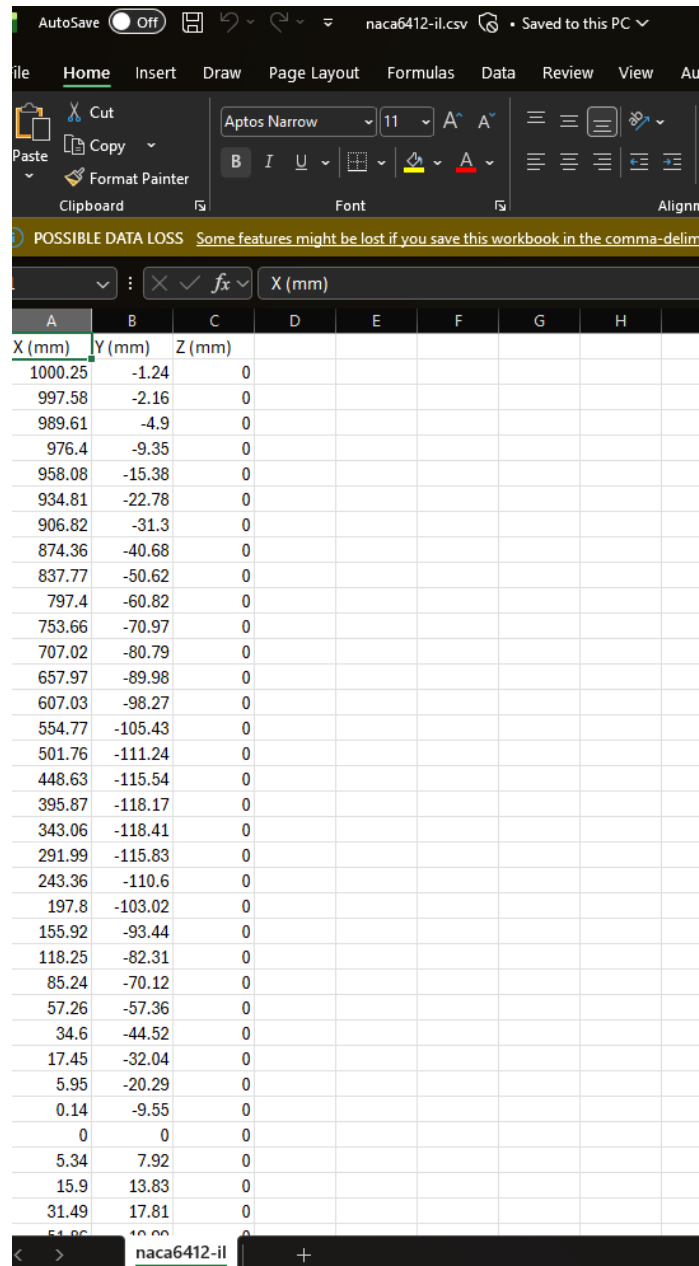
AUTOMATION PRINCIPLE

The project intentionally distinguishes implemented automation from future opportunities. Excel-to-CATIA macro-assisted geometry creation and Python post-processing were implemented. STAR-CCM+ Java automation is presented as a credible next development step, not as completed work.

4. Geometry Development

4.1 Coordinate Preparation

The source coordinate file contained 62 ordered points defined in millimetres. All z-coordinates were zero, placing the section on the global XY plane. The point sequence described the profile from the trailing-edge region around the leading edge and back to the trailing edge. The nominal chord was approximately 1,000 mm.



A	B	C	D	E	F	G	H	I
X (mm)	Y (mm)	Z (mm)						
1000.25	-1.24	0						
997.58	-2.16	0						
989.61	-4.9	0						
976.4	-9.35	0						
958.08	-15.38	0						
934.81	-22.78	0						
906.82	-31.3	0						
874.36	-40.68	0						
837.77	-50.62	0						
797.4	-60.82	0						
753.66	-70.97	0						
707.02	-80.79	0						
657.97	-89.98	0						
607.03	-98.27	0						
554.77	-105.43	0						
501.76	-111.24	0						
448.63	-115.54	0						
395.87	-118.17	0						
343.06	-118.41	0						
291.99	-115.83	0						
243.36	-110.6	0						
197.8	-103.02	0						
155.92	-93.44	0						
118.25	-82.31	0						
85.24	-70.12	0						
57.26	-57.36	0						
34.6	-44.52	0						
17.45	-32.04	0						
5.95	-20.29	0						
0.14	-9.55	0						
0	0	0						
5.34	7.92	0						
15.9	13.83	0						
31.49	17.81	0						
51.86	10.00	0						

Figure 4.1. Source NACA 6412 coordinate dataset prepared in millimetres for coordinate-driven geometry generation.

4.2 CATIA V5 Construction

The coordinates were imported using the CATIA GeometryFromExcel macro workflow. The macro read the prepared coordinate rows and created the corresponding points in Generative Shape Design. The imported point set defined the profile perimeter, which was converted into a closed section and extruded to form the source CAD geometry. The completed 1 m chord model was retained as a CATIA part and exported as a STEP file for STAR-CCM+. The source domain geometry included a nominal 5 m extrusion, while the aerodynamic solution itself used STAR-CCM+ two-dimensional meshing and therefore represented an infinite section rather than a finite-span wing.

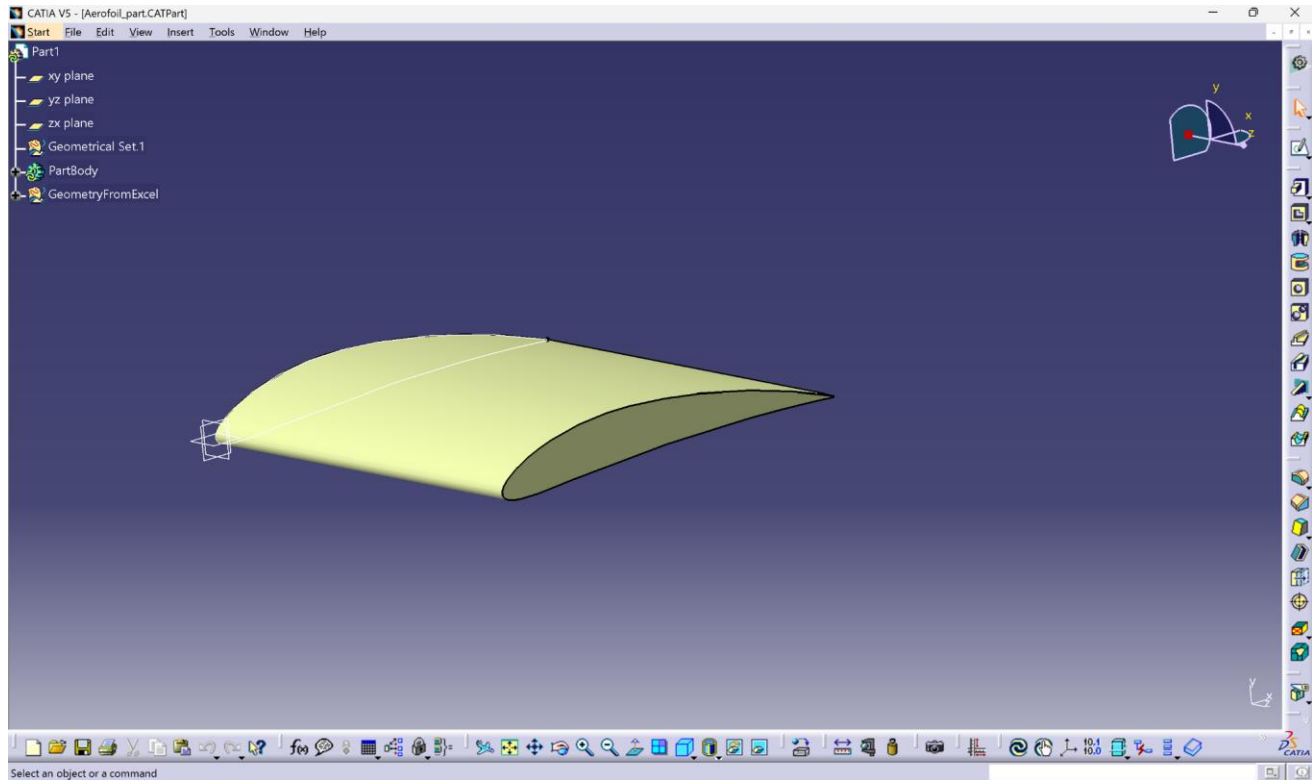


Figure 4.2. Coordinate-driven NACA 6412 geometry generated in CATIA V5 using the GeometryFromExcel workflow.

4.3 Geometry-Control Observations

The coordinate set retained a small finite trailing-edge thickness. XFOIL subsequently reported a normalised trailing-edge gap of approximately 0.00126c and poor local point distribution before repaneling. These features were retained as documented geometry limitations rather than silently corrected after simulation.

5. CFD Methodology

5.1 Operating Conditions

PARAMETER	VALUE
Aerofoil	Inverted NACA 6412
Chord	1.000 m
Freestream velocity	50.0 m/s
Air density	1.225 kg/m ³
Dynamic viscosity	1.789 × 10 ⁻⁵ Pa·s
Static pressure	101,325 Pa
Reynolds number	3.424 × 10 ⁶
Mach number	0.147
Dynamic pressure	1,531.25 Pa
Iterations per design	1,000

5.2 Domain and Boundary Conditions

The aerofoil lay on the global XY plane inside a 50 m × 50 m square far-field domain, equivalent to 25 chord lengths in each principal direction. Angle of attack was imposed by rotating the aerofoil about its leading-edge region while retaining the global freestream vector.

BOUNDARY / DEFINITION	SETTING
Upstream	Velocity inlet, (50, 0, 0) m/s
Upper far field	Velocity inlet, (50, 0, 0) m/s
Lower far field	Velocity inlet, (50, 0, 0) m/s
Downstream	Pressure outlet, 0 Pa gauge
Aerofoil	No-slip wall
Downforce direction	(0, 1, 0); negative result denotes downforce
Drag direction	(1, 0, 0)

In the two-dimensional region, the front, upper and lower domain faces were defined as velocity inlets with the common flow-aligned vector (50, 0, 0) m/s, while the rear face was a 0 Pa gauge pressure outlet. The 3.0 m far-field surface control was applied consistently to all four outer boundaries. The 25-chord clearance and coarse outer-boundary treatment concentrated cell density around the aerofoil and wake; a dedicated freestream or far-field boundary formulation remains a useful future sensitivity check.

5.3 Physics and Solver

CATEGORY	MODEL
Dimensionality	Two-dimensional
Time treatment	Steady-state
Flow solver	Segregated
Density treatment	Constant-density air
Turbulence treatment	RANS, k- ϵ [4]
Inlet turbulence	STAR-CCM+ standard/default values
k- ϵ variant	Not retained in available evidence
Wall treatment	Not retained in available evidence

5.4 Mesh Strategy

PARAMETER	MEDIUM-MESH SETTING
Base size	0.010 m = c/100
Target surface size	0.010 m
Minimum surface size	10% of base = 0.001 m
Surface growth rate	1.2
Surface-mesher size	50% of base = 0.005 m
Prism layers	5
Total prism thickness	0.010 m
Leading-edge control	Cylindrical volumetric refinement
Wake refinement length	4.0 m
Wake isotropic size	400% of base = 0.040 m
Far-field boundary surface control	Target and minimum surface sizes of 3.0 m, applied to the front, upper and lower velocity-inlet boundaries and the rear pressure-outlet boundary

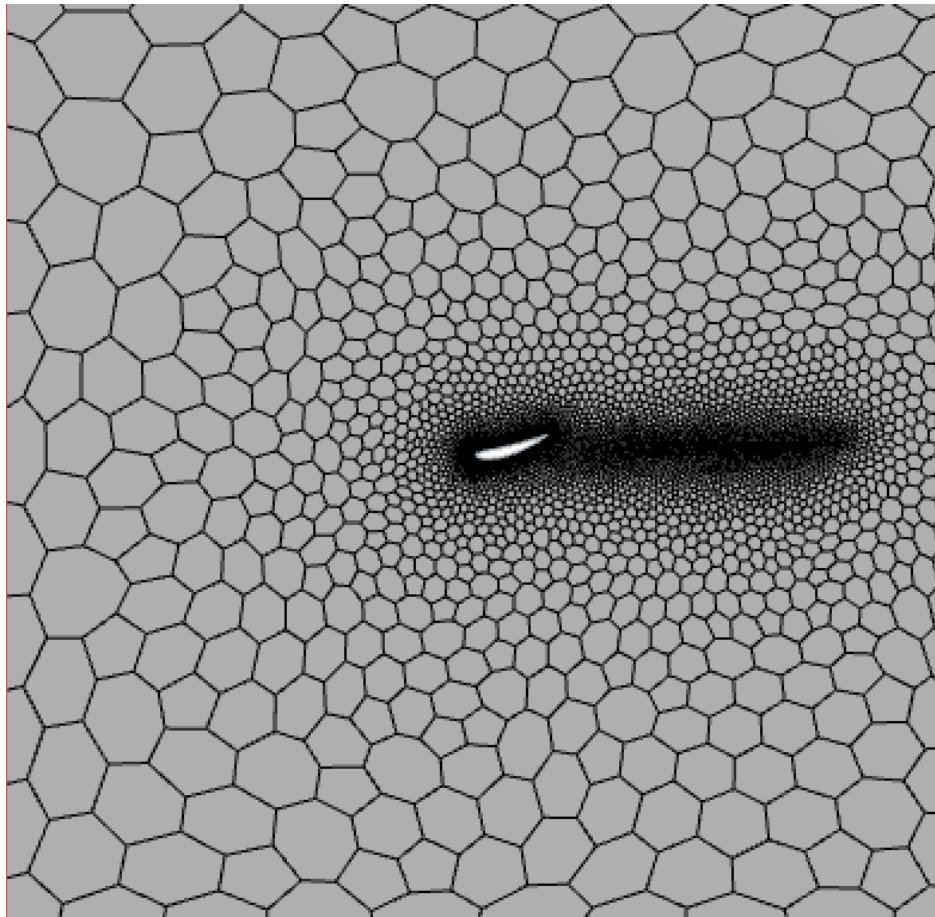


Figure 5.1. Domain-scale view of the selected medium mesh, showing local aerofoil refinement and a refined downstream wake.

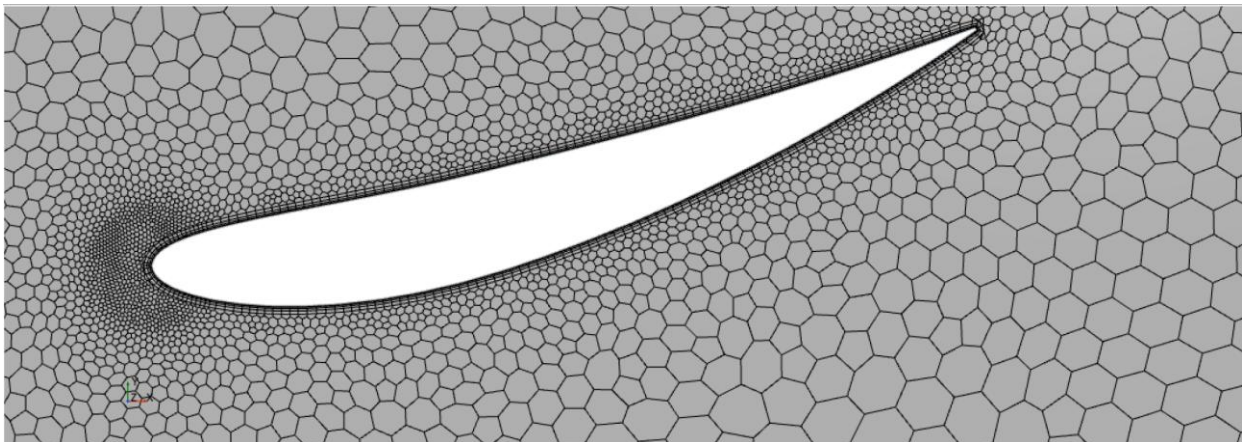


Figure 5.2. Near-field view of the selected medium mesh, showing leading-edge and wall-adjacent refinement.

A dedicated surface control was applied to the four far-field boundaries of the two-dimensional domain. The front, upper and lower faces were velocity inlets and the rear face was the pressure outlet. Both the target and minimum surface sizes were set to 3.0 m on these boundaries. This control intentionally maintained a coarse far-field mesh, while the global base size and the local aerofoil, leading-edge and wake controls provided the higher resolution required in the aerodynamic near field.

5.5 Reference-Area Correction

The original STAR-CCM+ lift and drag coefficient reports used different directional reference areas. To enable credible comparison, Python preserved the dimensional forces and recalculated both coefficients using a common 1.000 m^2 area. The corrected coefficients, rather than the original on-screen coefficients, form the primary dataset in Sections 7 to 10.

6. Mesh Sensitivity and Selection

MESH	BASE SIZE	CELLS	FACES	VERTICES
Coarse	0.100 m	833	2,207	1,525
Medium	0.010 m	6,662	18,597	12,210
Fine	0.001 m	190,488	558,144	369,924

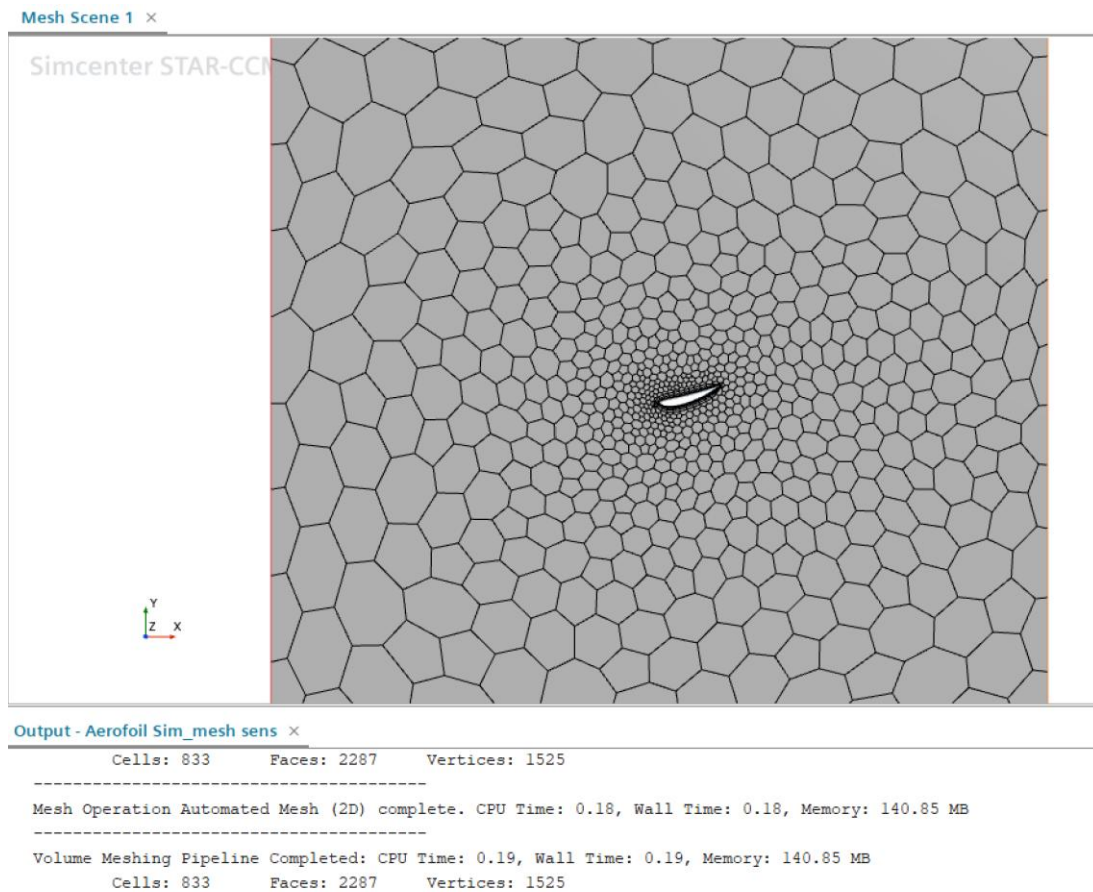


Figure 6.1. Coarse mesh using a 0.100 m base size.

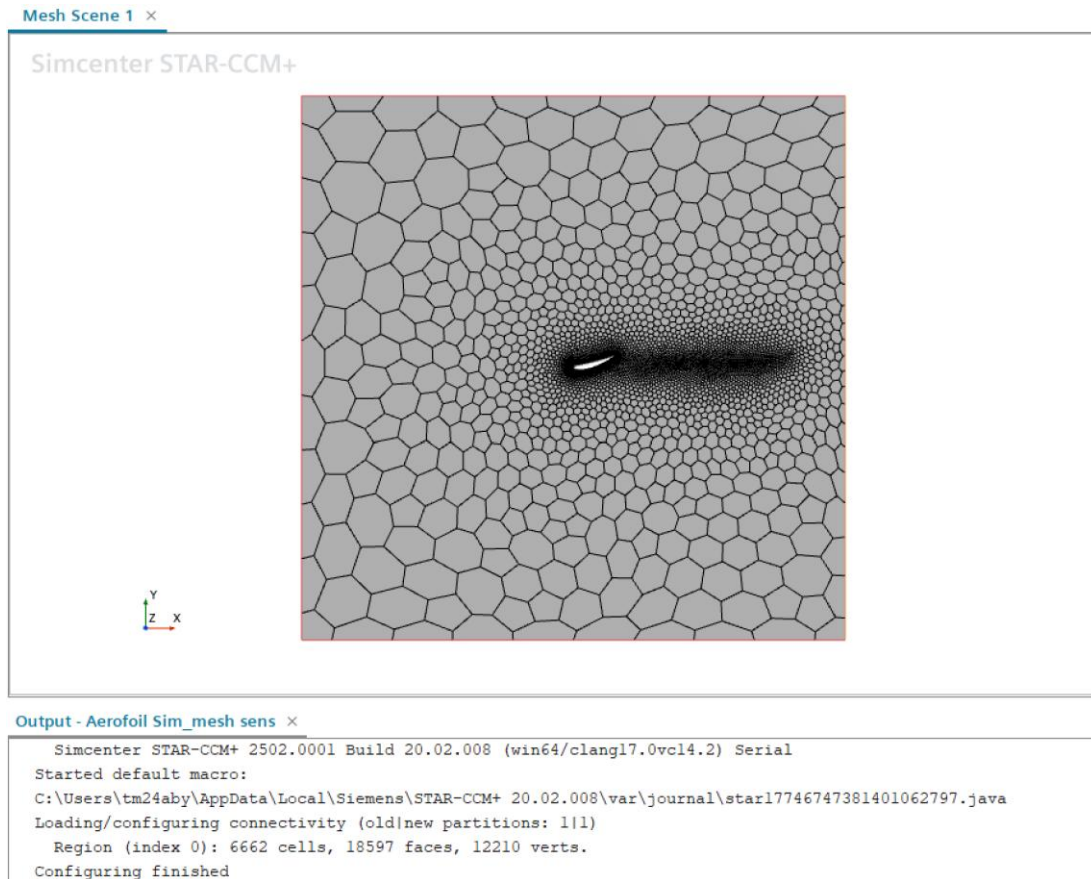


Figure 6.2. Medium mesh using a 0.010 m base size.

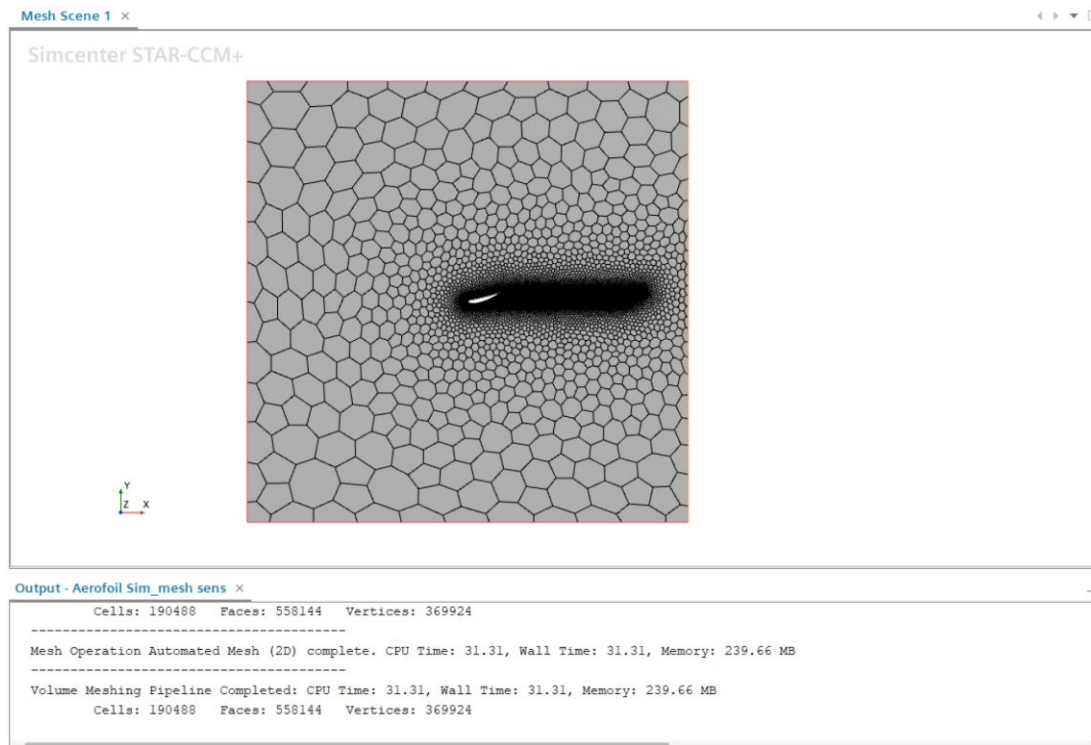


Figure 6.3. Fine mesh using a 0.001 m base size.

6.1 Convergence Behaviour

MESH	RAW DISPLAYED CL	RAW DISPLAYED CD	ASSESSMENT
Coarse	-0.9524	0.7046	Persistent periodic oscillation; rejected
Medium	-1.8577	0.2192	Stable late-iteration force monitors
Fine	-1.6652	0.2158	Low-frequency force variation remains

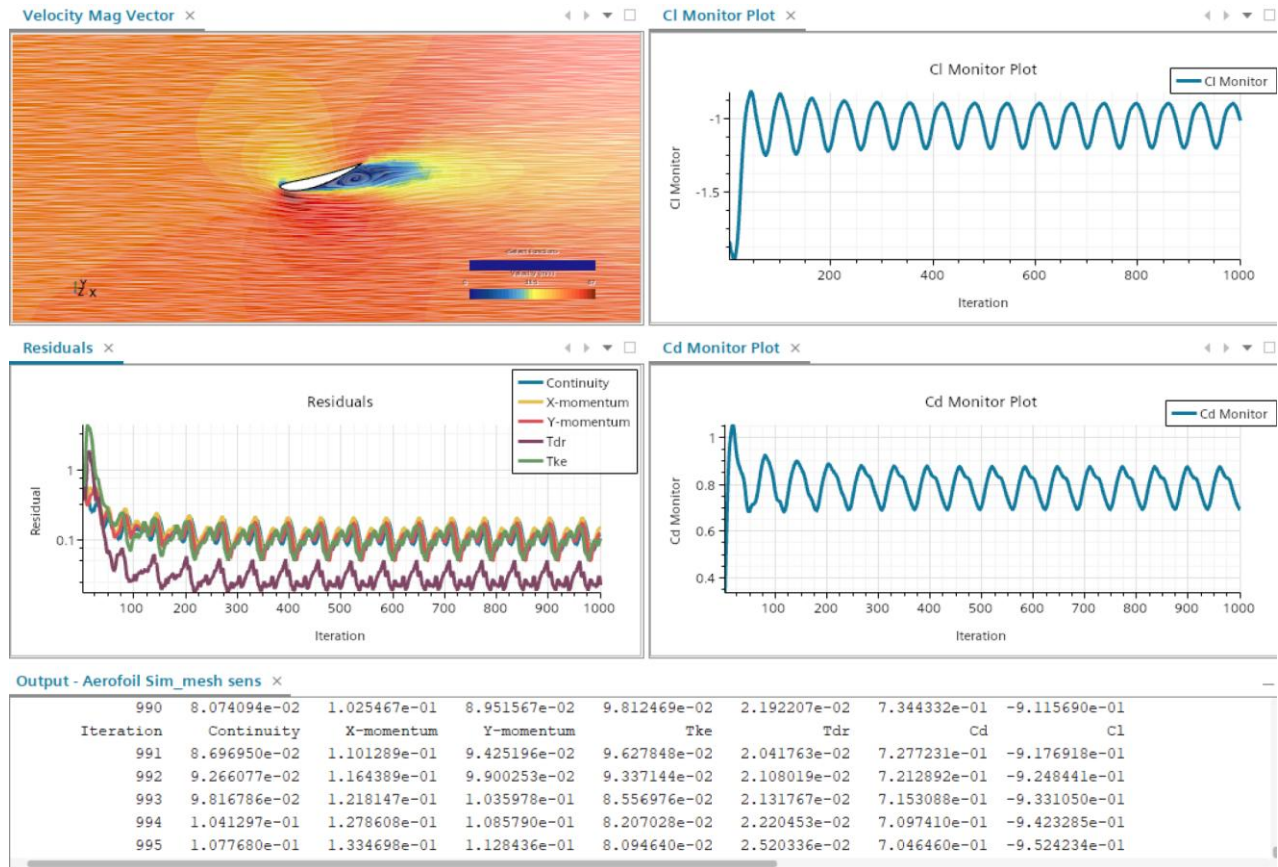


Figure 6.4. Coarse-mesh convergence history showing persistent residual and force-coefficient oscillation.

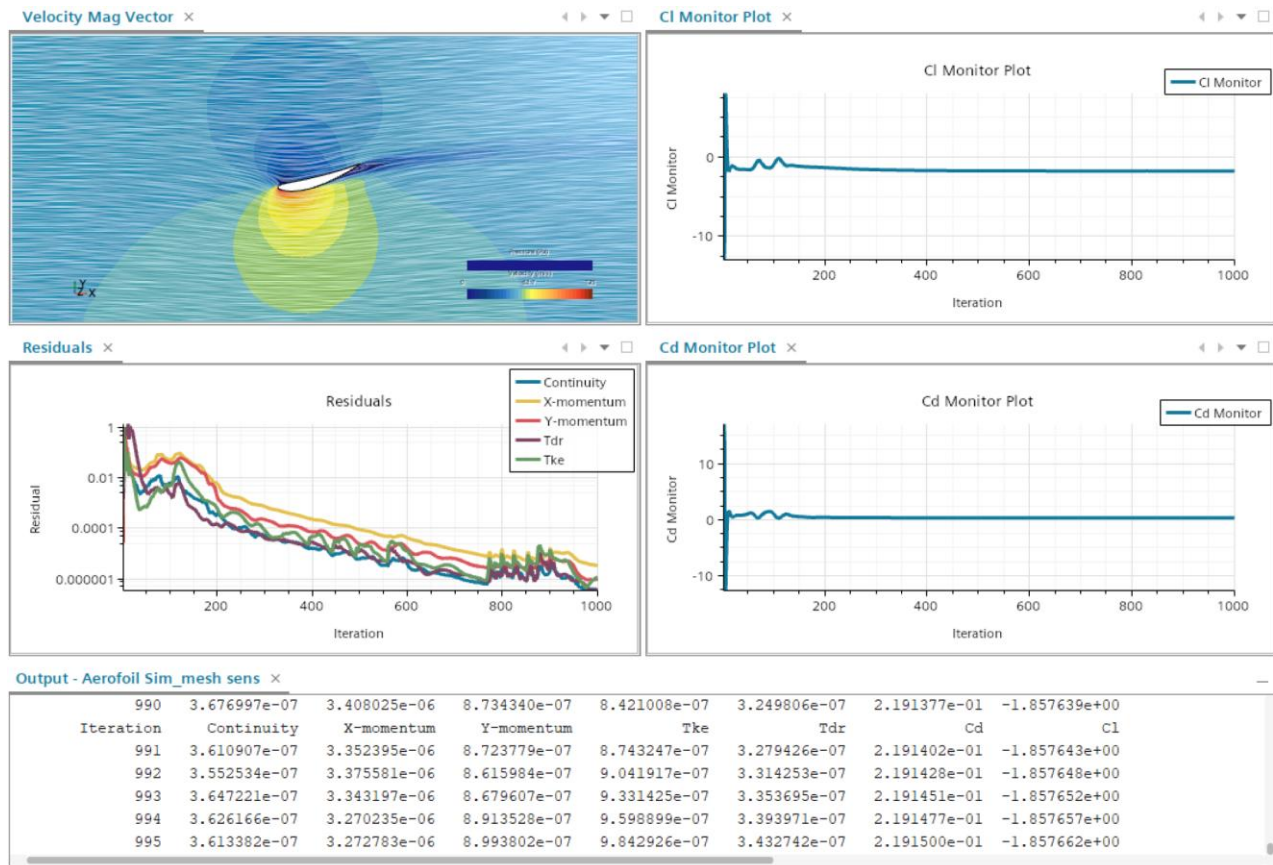


Figure 6.5. Medium-mesh convergence history showing stable late-iteration force coefficients.

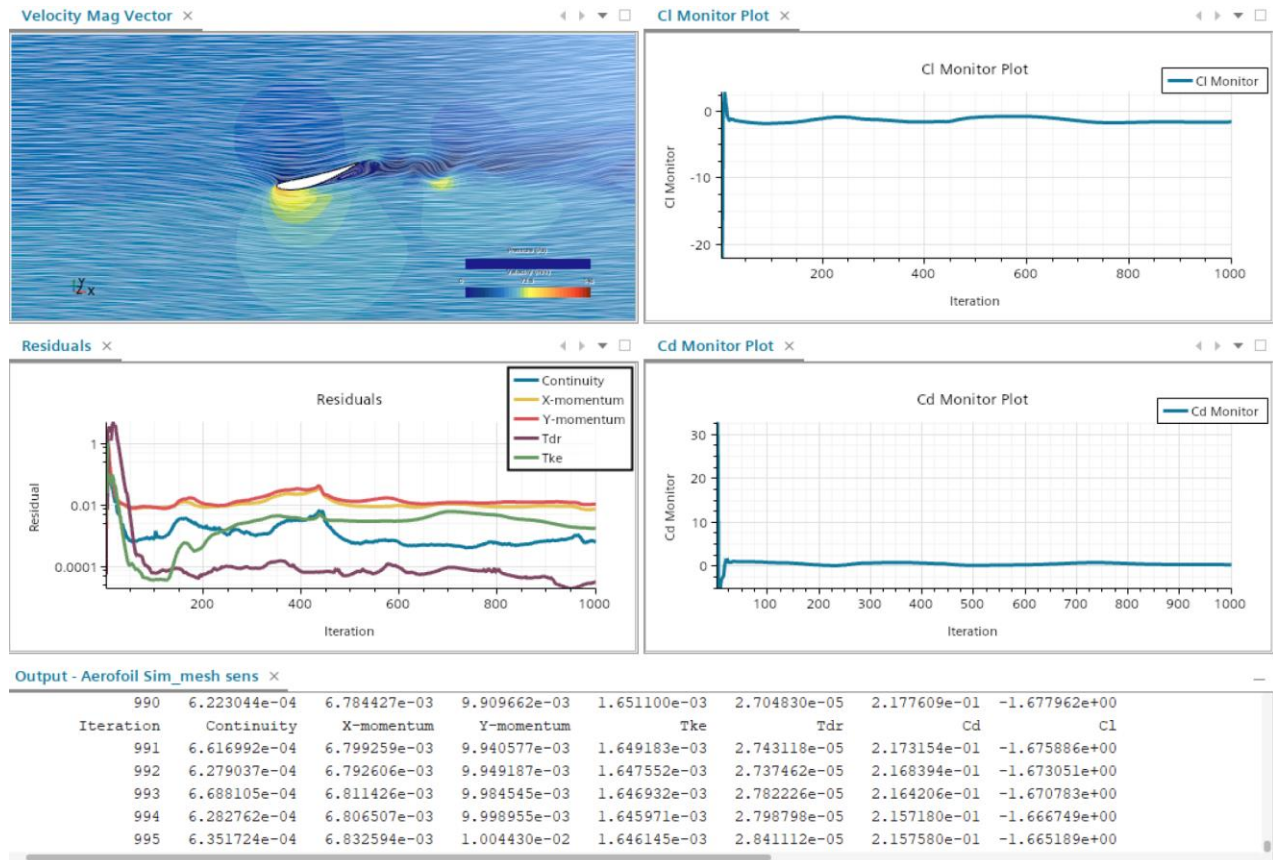


Figure 6.6. Fine-mesh convergence history showing remaining low-frequency force variation after 1,000 iterations.

6.2 Selection Decision

The coarse discretisation was under-resolved and did not establish a credible steady solution. Relative to the medium mesh, the fine mesh changed raw displayed drag by approximately 1.55%, but changed lift magnitude by approximately 10.4% and retained low-frequency oscillation. Strict lift mesh independence was therefore not demonstrated.

SELECTED WORKING MESH

The 6,662-cell medium mesh was selected for the angle-of-attack sweep because it achieved the strongest steady-state convergence at materially lower computational cost. The medium-to-fine lift sensitivity is retained as a quantified model limitation, not hidden by the selection decision.

7. Angle-of-Attack Results

7.1 Corrected Performance Dataset

Nine operating points were evaluated for 1,000 iterations. The initial 0°, 5°, 10°, 15° and 20° sweep established the global trend; 16° to 19° cases refined the peak-loading region. Corrected values use the common 1.000 m² reference area.

AOA	CL	CD	CL /CD	DOWNFORCE N/m	DRAG N/m
0	-0.6135	0.0303	20.27	939.4	46.3
5	-1.1035	0.0356	31.04	1,689.8	54.4
10	-1.5443	0.0442	34.93	2,364.6	67.7
15	-1.7859	0.0635	28.13	2,734.6	97.2
16	-1.7863	0.0709	25.19	2,735.3	108.6
17	-1.7400	0.0797	21.84	2,664.4	122.0
18	-1.6920	0.0919	18.42	2,590.9	140.7
19	-1.6008	0.1043	15.34	2,451.2	159.8
20	-1.5100	0.1194	12.65	2,312.2	182.8

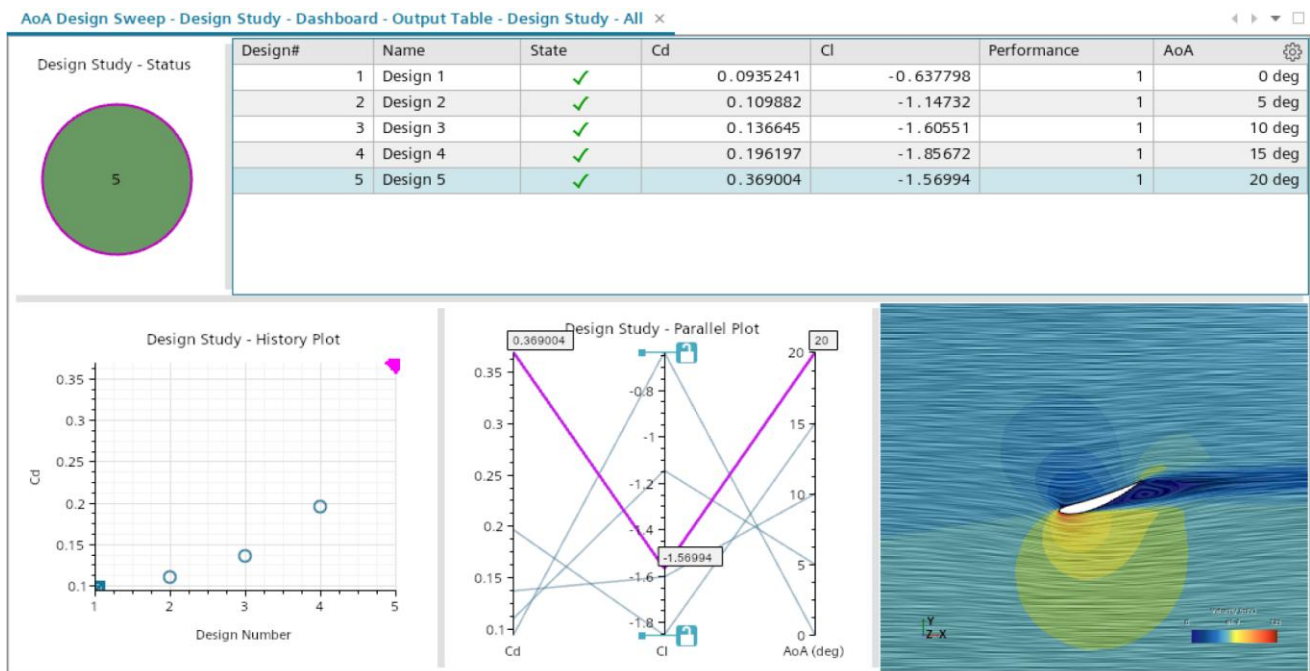


Figure 7.1. STAR-CCM+ angle-of-attack sweep study evidence.

7.2 Loading Trend

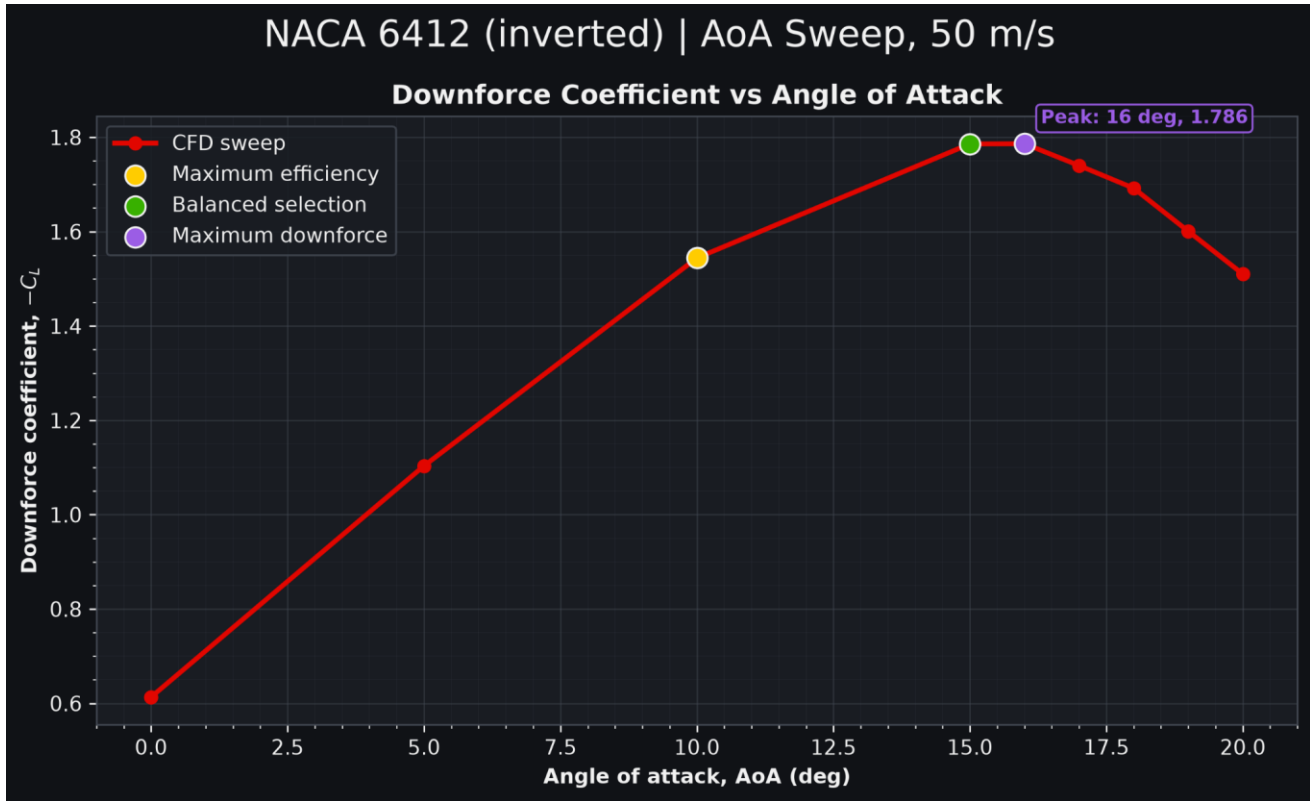


Figure 7.2. Corrected downforce coefficient versus angle of attack.

Downforce increased from 0° to a numerical maximum at 16°. The 15° and 16° coefficients were -1.78588 and -1.78634 respectively, a downforce gain of only 0.026% at 16°. Beyond 16°, downforce declined continuously, providing clear evidence of post-peak loading behaviour within the steady model.

7.3 Drag Trend

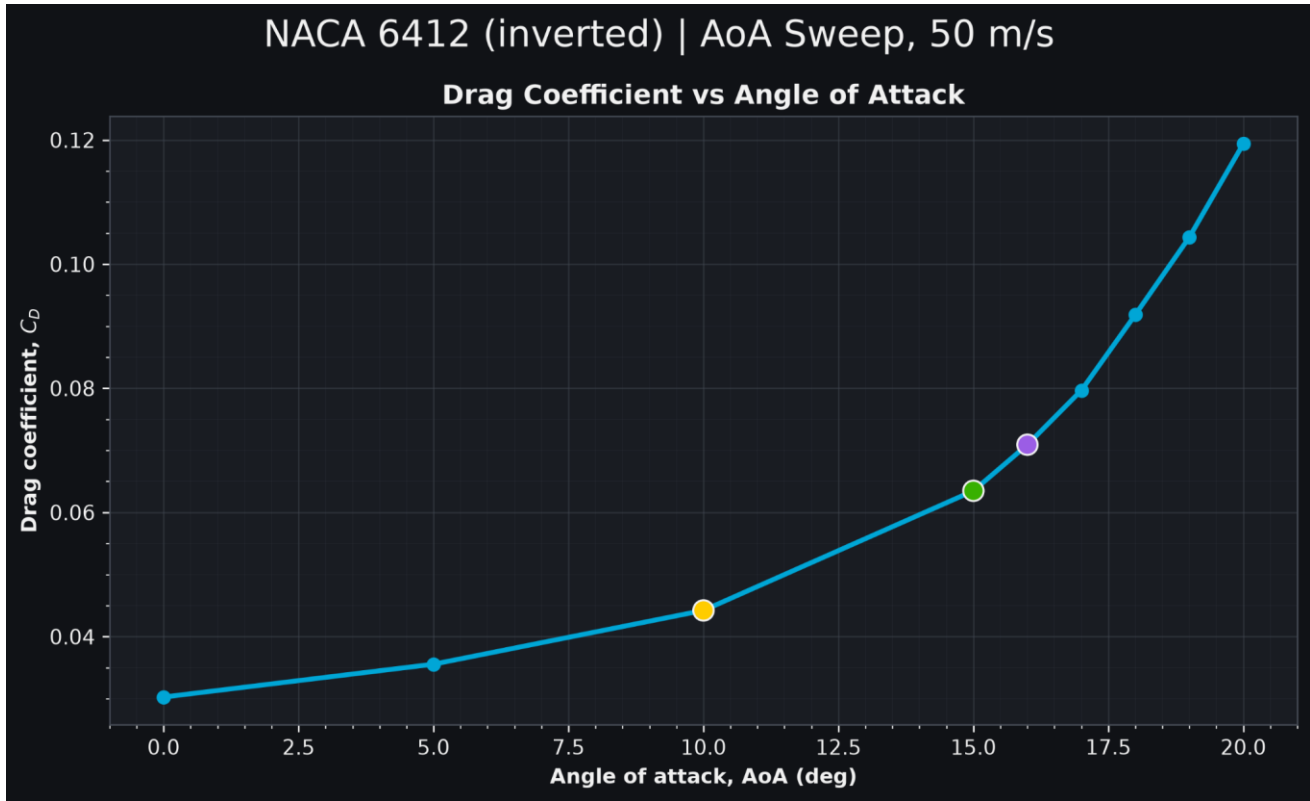


Figure 7.3. Corrected drag coefficient versus angle of attack.

Drag increased monotonically across the sweep. C_D rose from 0.06348 at 15° to 0.07091 at 16°, an 11.70% increase. At 20°, C_D reached 0.11940 while downforce had fallen to 1.5100, confirming that the highest-incidence cases carried a substantial performance penalty.

7.4 Aerodynamic Efficiency

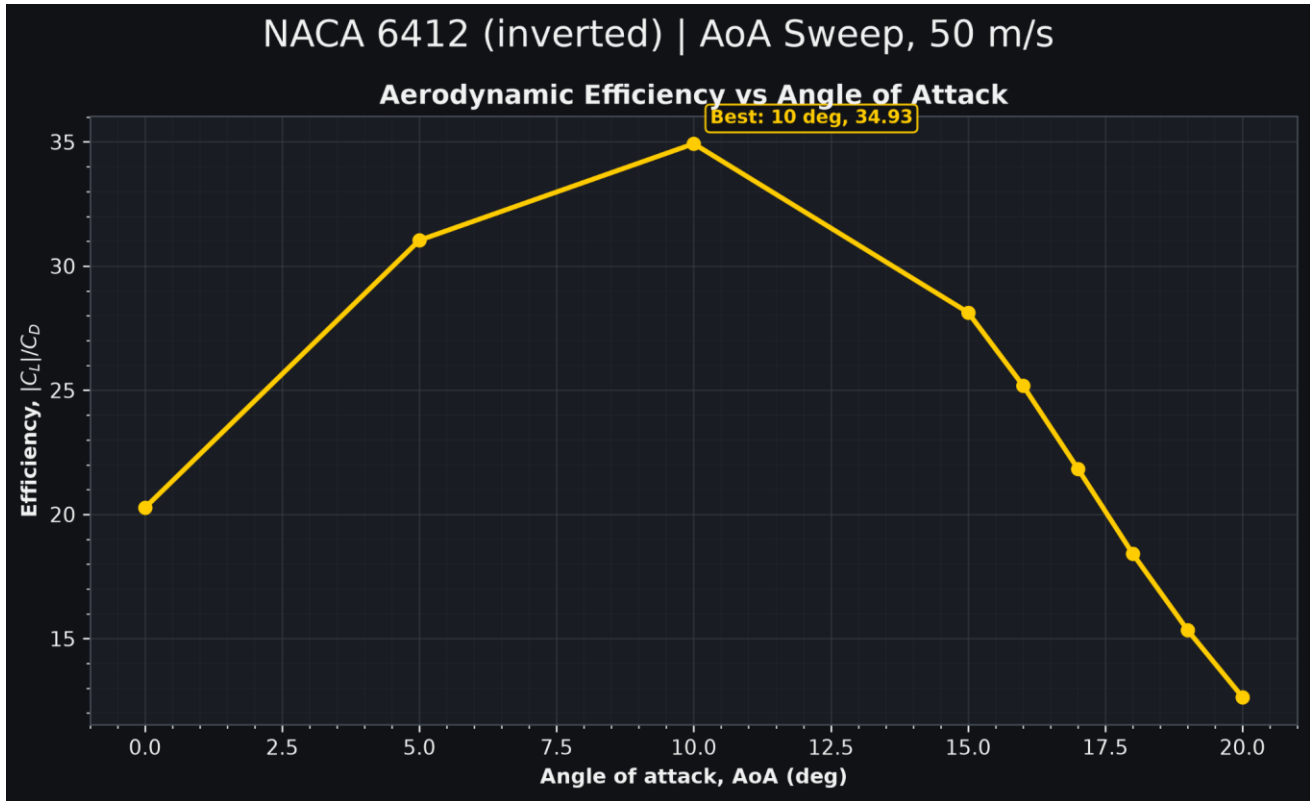


Figure 7.4. Aerodynamic efficiency, $|C_L|/C_D$, versus angle of attack.

Efficiency peaked at 10° with $|C_L|/C_D = 34.93$. The 15° balanced point retained $|C_L|/C_D = 28.13$ while producing 15.65% more downforce than 10°. The 16° maximum-downforce case reduced efficiency to 25.19.

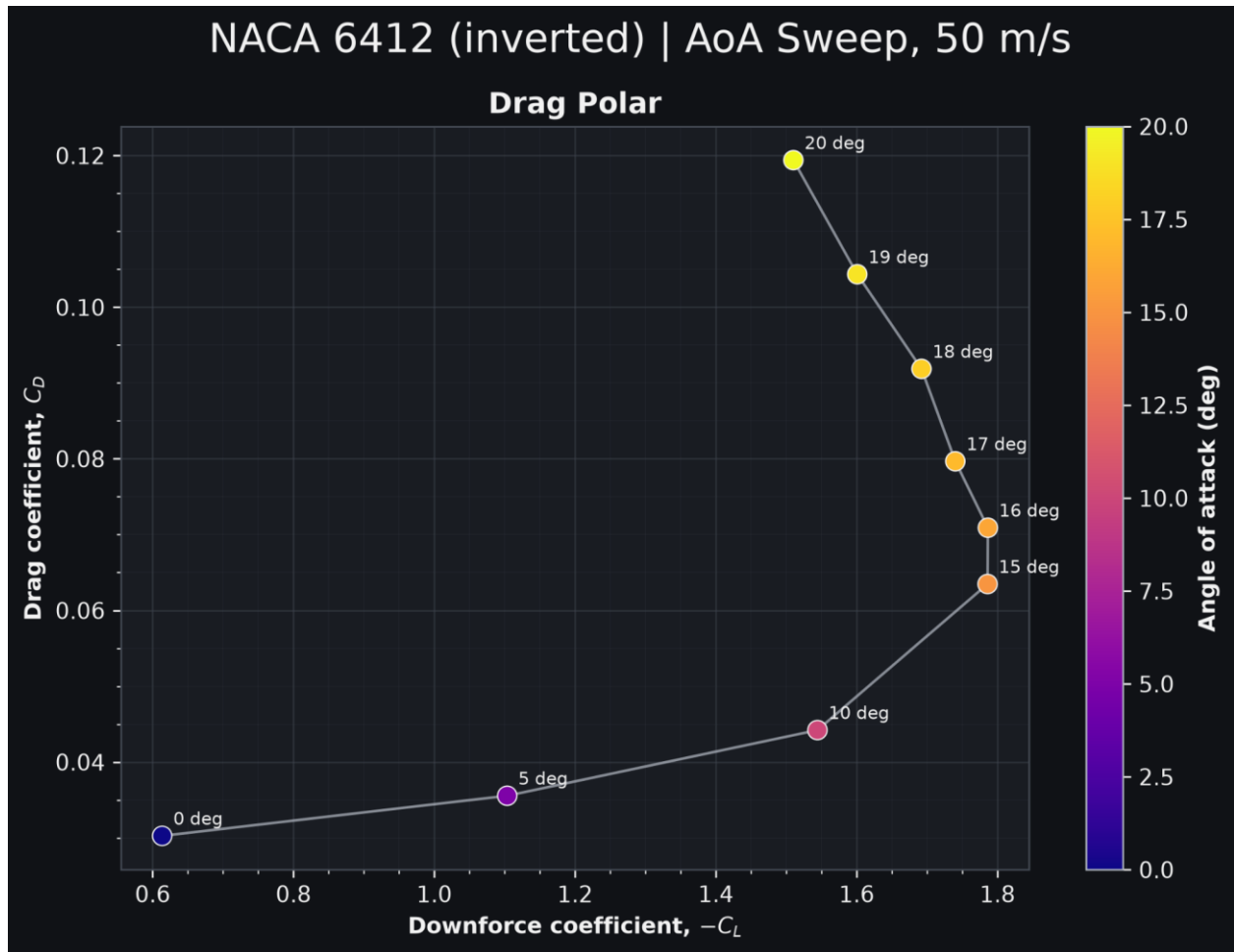


Figure 7.5. Corrected drag polar for the CFD design sweep.

7.5 Dimensional Loading and Flow Field

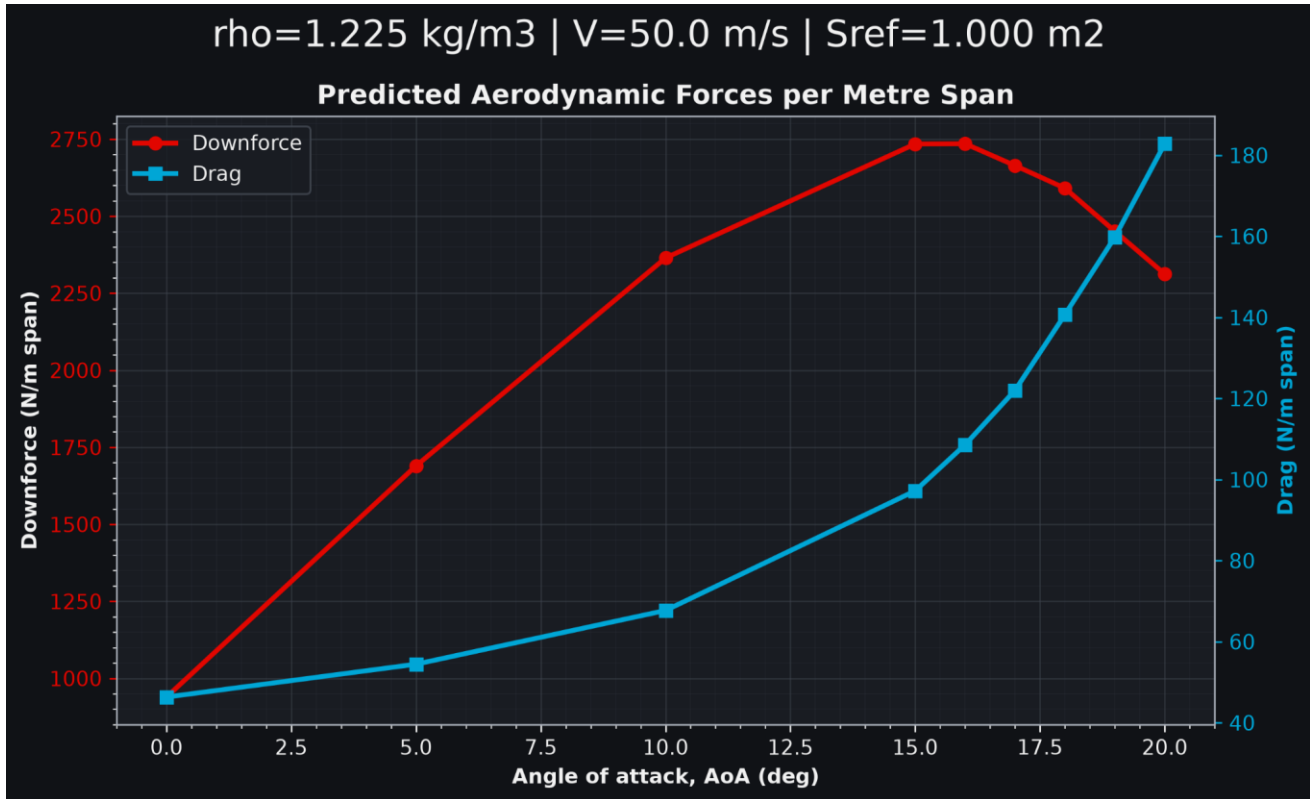


Figure 7.6. Downforce and drag per metre span at the declared reference conditions.



Figure 7.7. Static-pressure field at 16°, the computed maximum-downforce operating point.

At 16°, the corrected section loads were approximately 2,735.3 N/m downforce and 108.6 N/m drag. The pressure field shows a strong differential across the inverted section and concentrated leading-edge loading. The post-16° reduction in downforce, combined with rising drag, is consistent with progressive separation.

8. Python Post-Processing and Traceability

A dedicated Python workflow converted raw exports into auditable engineering outputs. The script imported the corrected lift and drag files, checked for missing and duplicate values, confirmed matching AoA sets, computed downforce magnitude, efficiency and dimensional forces, and generated publication-quality figures.

CONTROL	IMPLEMENTATION
Input QA	Missing, duplicate and non-numeric value checks
Dataset alignment	Lift and drag AoA matching
Coefficient convention	Negative CL converted to positive downforce magnitude for comparison
Force calculation	$q\infty S_{ref}$ applied to corrected coefficients
Reference audit	Original and 1 m ² coefficient definitions checked
Verification	XFOIL polar parsed and common angles matched
Metrics	MAE, RMSE, MAPE and gradient difference
Traceability	Timestamps, software versions and SHA-256 input hashes
Outputs	PNG, SVG, CSV, TXT, JSON and interactive HTML report

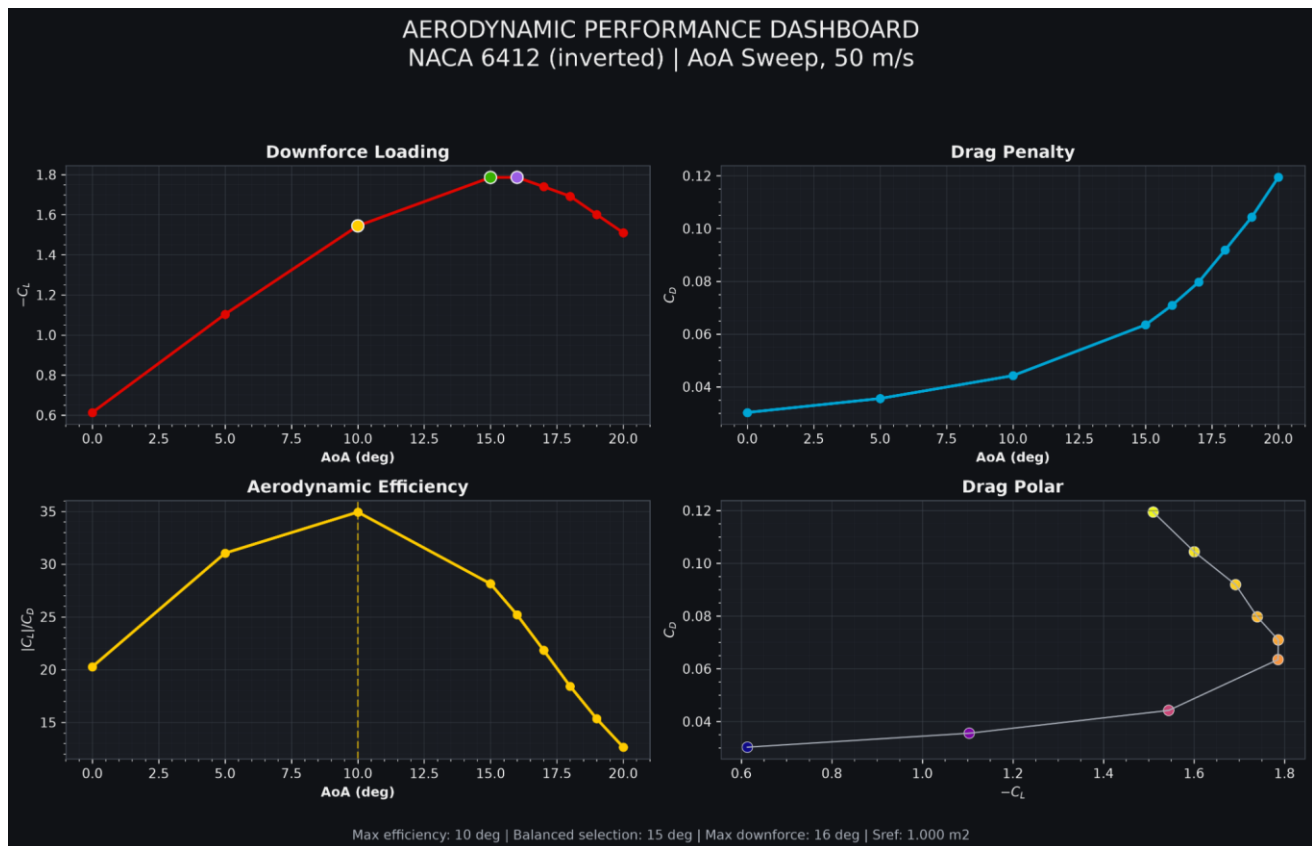


Figure 8.1. Automated aerodynamic performance dashboard generated by the final Python workflow.

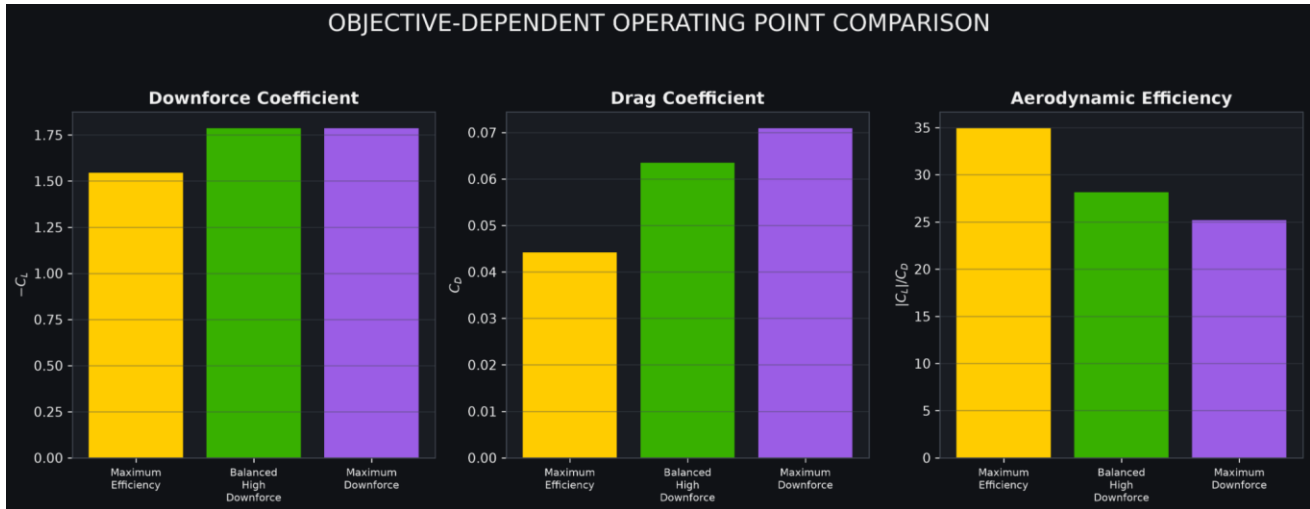


Figure 8.2. Automated comparison of the efficiency, balanced and maximum-downforce operating points.

CONFIGURATION DISCIPLINE

The Python workflow separates raw solver reporting from corrected portfolio reporting while preserving dimensional forces. This provides a reproducible audit trail for the reference-area correction and the final engineering decision.

9. XFOIL Numerical Verification

9.1 Method

XFOIL 6.99 was run as an independent viscous airfoil analysis at Reynolds number 3.423×10^6 and Mach number 0.147. The 62-point coordinate input generated warnings concerning local point distribution near the trailing edge. The PANE command was used to produce 160 computational nodes. Free transition was retained with $N_{crit} = 9.0$. The polar covered 0° to 20° in nominal 0.5° increments. [2]

The 15° XFOIL case did not converge and was excluded rather than interpolated. The saved polar contained duplicate angle rows after an append operation; the Python parser retained the last row for each angle before matching common CFD and XFOIL points. XFOIL is a lower-order numerical method, so this comparison is described as independent numerical verification rather than experimental validation.

9.2 Verification Metrics

METRIC	VALUE
Common matched points	8
CL RMSE	0.1879
CL MAE	0.1678
CL mean absolute percentage error	10.15%
CD RMSE	0.02171
CD MAE	0.02112
CD mean absolute percentage error	139.05%
CFD peak-loading AoA	16°
XFOIL peak-loading AoA	16°
CFD loading slope	0.09308 per degree
XFOIL lift slope	0.10018 per degree
Slope difference	-7.09%

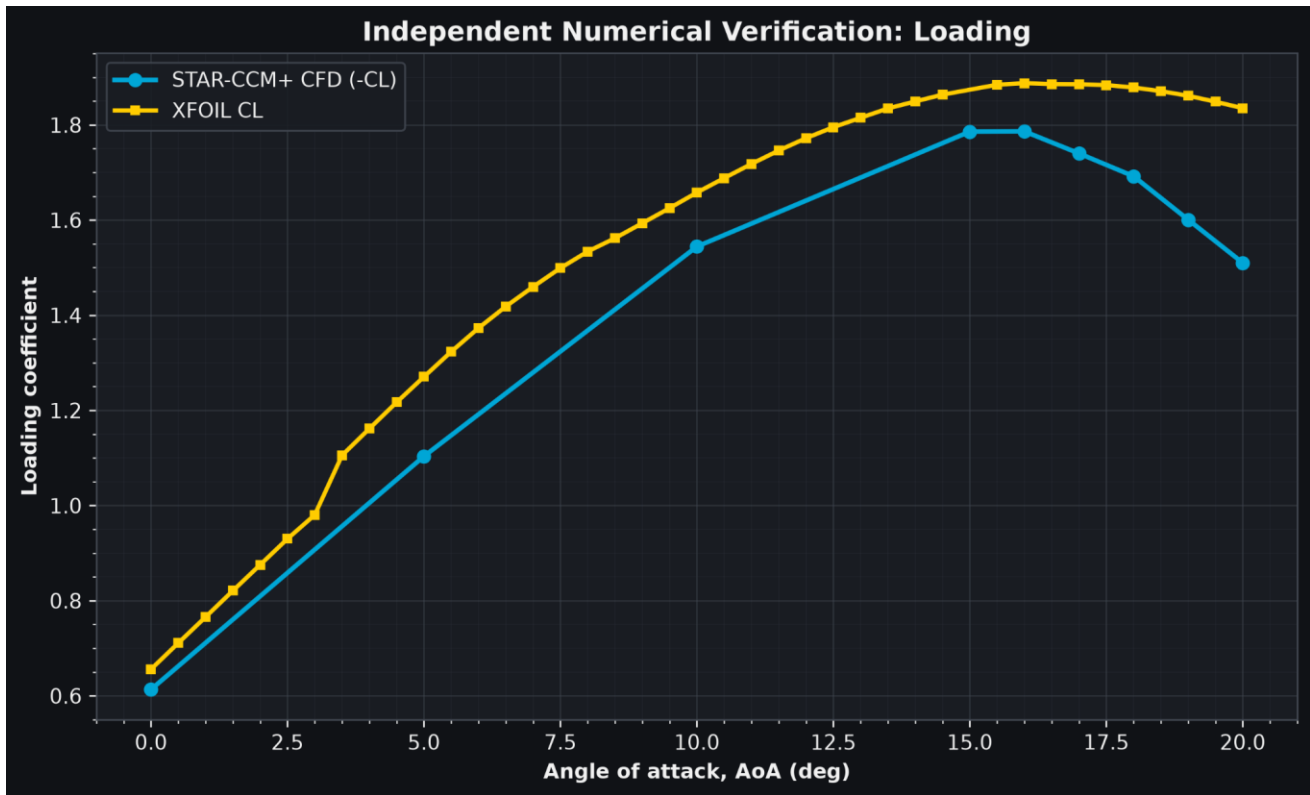


Figure 9.1. CFD and XFOIL loading comparison at common converged operating points.

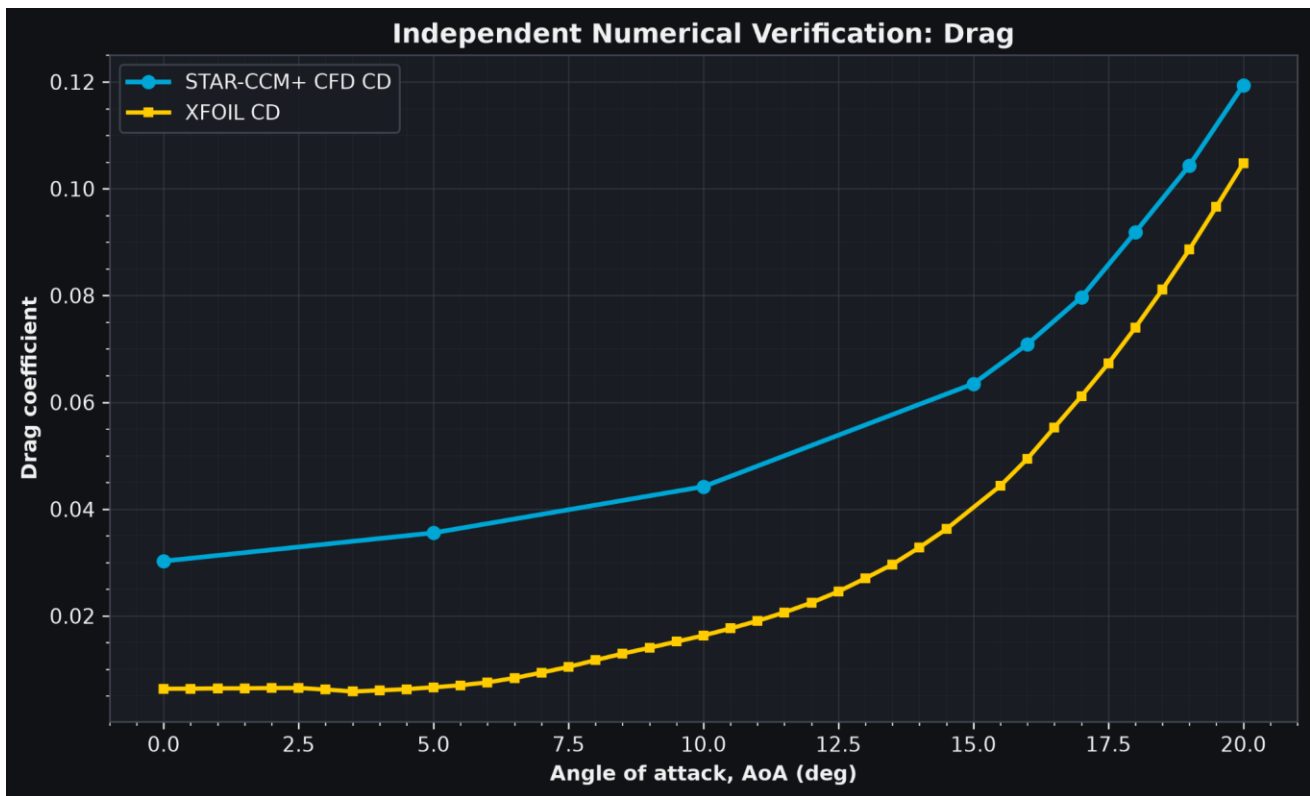


Figure 9.2. CFD and XFOIL drag comparison at common converged operating points.

9.3 Interpretation

Both methods identified 16° as the peak-loading angle, and the linear loading-gradient difference was 7.09%. This provides useful trend-level support for the CFD loading result. STAR-CCM+ predicted lower loading than XFOIL at every matched point, with differences increasing beyond the peak.

Drag agreement was materially weaker. At 0° , CFD predicted $CD = 0.03026$ compared with 0.00636 from XFOIL. The percentage discrepancy reduced at high incidence, reaching approximately 13.95% at 20° . Differences are expected because steady RANS with the documented mesh and turbulence treatment is not numerically equivalent to XFOIL's coupled panel and integral boundary-layer formulation, particularly near separation.

10. Engineering Discussion and Recommendation

10.1 Objective-Dependent Operating Points

OBJECTIVE	SELECTED AOA	DOWNFORCE COEFF.	DRAW COEFF.	EFFICIENCY
Maximum efficiency	10°	1.5443	0.0442	34.93
Balanced high downforce	15°	1.7859	0.0635	28.13
Maximum downforce	16°	1.7863	0.0709	25.19

10.2 Balanced-Point Trade-off

Moving from 15° to 16° increased downforce coefficient by only 0.00046, equivalent to approximately 0.7 N/m at the declared condition. Over the same change, drag increased by approximately 11.4 N/m. Expressed relative to the 16° case, selecting 15° retains 99.97% of maximum downforce while reducing drag by 10.47%.

FINAL RECOMMENDATION

Use 15° as the recommended balanced high-downforce operating point for this isolated two-dimensional section study. Use 10° where aerodynamic efficiency is the dominant requirement, and reserve 16° only for a maximum-loading objective where the additional drag and reduced separation margin are acceptable.

10.3 Formula 1-Focused Interpretation

A Formula 1 aerodynamic device is rarely selected on peak section loading alone. The useful operating point must tolerate ride-height variation, pitch, yaw, upstream-wheel wake, downstream interaction and regulation-constrained packaging. The 15° choice reflects that systems-level mindset: it avoids paying a large drag penalty for a negligible isolated-section loading gain and retains greater margin before the measured post-peak decline.

The 10° efficiency point may be more appropriate for low-drag circuit requirements or for a wing operating within a larger multi-element system where downstream elements amplify total loading. The 16° point may suit a local maximum-load target, but it is more sensitive to modelling uncertainty because it lies at the computed peak and adjacent to separated-flow behaviour.

10.4 Decision Confidence

EVIDENCE	CONFIDENCE / EFFECT
AoA trend	High within the computed dataset; monotonic trends are clear
Peak location	Moderate; CFD and XFOIL both identify 16°
Absolute drag	Low to moderate; strong CFD–XFOIL discrepancy
Mesh independence	Not demonstrated for lift
15° versus 16° decision	Strong on trade-off basis; loading difference is negligible
Transfer to a complete car	Low without 3D interaction and experimental evidence

11. Limitations and Future Development

LIMITATION	ENGINEERING CONSEQUENCE	RECOMMENDED ACTION
Two-dimensional isolated section	No finite-span, endplate or vehicle-interaction physics	Build a 3D half-car or subsystem model
Steady RANS near stall	May suppress or misrepresent unsteady separation	Run transient URANS at 15°–20°
k-ε variant and wall treatment unrecorded	Reduces reproducibility and confidence	Archive continuum screenshots and exported settings
No y+ evidence	Near-wall suitability cannot be verified	Generate y+ reports and design first-layer height
Lift mesh sensitivity ≈10.4%	Strict mesh independence not demonstrated	Add intermediate meshes and GCI-style assessment
Flow-aligned velocity top/bottom boundaries	Less general than a validated far-field condition	Use freestream/far-field boundaries and sensitivity test
Coarse 3.0 m far-field boundary surface control	Far-field resolution was intentionally much coarser than the local aerofoil and wake regions	Confirm boundary-distance and outer-mesh sensitivity in a future model
XFOIL is numerical comparison	Does not establish physical validation	Compare with wind-tunnel or published experimental data
Finite trailing-edge gap	Geometry differs from an ideal sharp profile	Quantify geometry sensitivity
Single speed and Re	Recommendation may not scale across car speed range	Run Reynolds-number and velocity sweeps

11.1 Motorsport-Focused Development Roadmap

- Create a transient URANS refinement around 14° to 18° with force averaging and spectral assessment.
- Replace the generic section with a parameterised motorsport profile and optimise camber, thickness, leading-edge radius and flap gap.
- Extend to a multi-element wing with endplates and evaluate interaction losses.
- Introduce ground proximity, ride-height and pitch sweeps to assess platform sensitivity.
- Couple aerodynamic loads into structural FEA to check deflection and load-path robustness.
- Establish experimental correlation using a scaled wind-tunnel model or validated public dataset.

12. Conclusion

A complete coordinate-to-decision workflow was developed for an inverted NACA 6412 aerofoil at 50 m/s. The study integrated CATIA geometry generation, STAR-CCM+ steady RANS, practical mesh sensitivity, a nine-point angle-of-attack sweep, reference-area correction, automated Python post-processing and XFOIL numerical verification.

The medium mesh was selected because it delivered the strongest steady-state convergence and enabled an efficient design sweep, although a 10.4% medium-to-fine lift difference prevented a strict mesh-independence claim. Downforce peaked at 16° with $CL = -1.7863$, while aerodynamic efficiency peaked at 10° with $|CL|/CD = 34.93$. XFOIL independently identified the same 16° peak-loading angle and produced a loading-gradient difference of -7.09%, providing trend-level support while highlighting weaker drag agreement.

The final engineering recommendation is 15° for balanced high-downforce operation. This point retains 99.97% of the computed maximum downforce while reducing drag by 10.47% relative to 16°. The recommendation is therefore based on performance robustness and trade-off quality rather than peak coefficient alone, consistent with a Formula 1-focused engineering decision process.

PORTFOLIO OUTCOME

The strongest evidence produced by this project is not a single coefficient. It is the controlled workflow linking geometry, numerical method, data quality, validation, transparent limitations and an objective-led design decision.

13. References

- [1] Airfoil Tools, “NACA 6412 (naca6412-il): geometry and coordinate data,” accessed 24 September 2026.
<http://airfoiltools.com/airfoil/details?airfoil=naca6412-il>
- [2] M. Drela and H. Youngren, “XFOIL 6.9 User Primer: Subsonic Airfoil Analysis and Design,” MIT Aero & Astro / Aircraft Inc., last documented update 30 November 2001. <https://v0xnihili.github.io/xfoil-docs/>
- [3] Siemens Digital Industries Software, “Simcenter STAR-CCM+ multiphysics solutions,” product fact sheet.
<https://static.sw.cdn.siemens.com/siemens-disw-assets/public/7m97PQE0esmEZd9JgDfzDb/en-US/siemens-sw-star-ccm-multiphysics-solutions-fact-sheet.pdf>
- [4] Siemens Xcelerator Academy, “Turbulence and turbulence modeling in Simcenter STAR-CCM+,” course ILT030006, accessed 24 September 2026.
- [5] Project evidence: naca6412-il.csv; STAR-CCM+ screenshots and corrected coefficient exports; Airfoil_F1_Post_Processing_Final.py; XFOIL_Run_Log.txt; XFOIL polar and validation CSV outputs.

Appendix A. Detailed Result Data

A.1 Corrected AoA Dataset

AOA	CL	CD	CL /CD	DOWNFORCE N/m	DRAW N/m
0	-0.6135	0.0303	20.27	939.4	46.3
5	-1.1035	0.0356	31.04	1,689.8	54.4
10	-1.5443	0.0442	34.93	2,364.6	67.7
15	-1.7859	0.0635	28.13	2,734.6	97.2
16	-1.7863	0.0709	25.19	2,735.3	108.6
17	-1.7400	0.0797	21.84	2,664.4	122.0
18	-1.6920	0.0919	18.42	2,590.9	140.7
19	-1.6008	0.1043	15.34	2,451.2	159.8
20	-1.5100	0.1194	12.65	2,312.2	182.8

A.2 CFD–XFOIL Common-Point Comparison

AOA	CFD LOAD	XFOIL CL	CL ERROR	CFD CD	XFOIL CD	CD ERROR
0	0.6135	0.6560	6.48%	0.0303	0.0064	375.80%
5	1.1035	1.2706	13.15%	0.0356	0.0066	436.26%
10	1.5443	1.6578	6.85%	0.0442	0.0163	170.75%
16	1.7863	1.8877	5.37%	0.0709	0.0494	43.45%
17	1.7400	1.8855	7.72%	0.0797	0.0612	30.29%
18	1.6920	1.8786	9.93%	0.0919	0.0740	24.14%
19	1.6008	1.8612	13.99%	0.1043	0.0886	17.76%
20	1.5100	1.8351	17.71%	0.1194	0.1048	13.95%

Appendix B. Supplementary Evidence

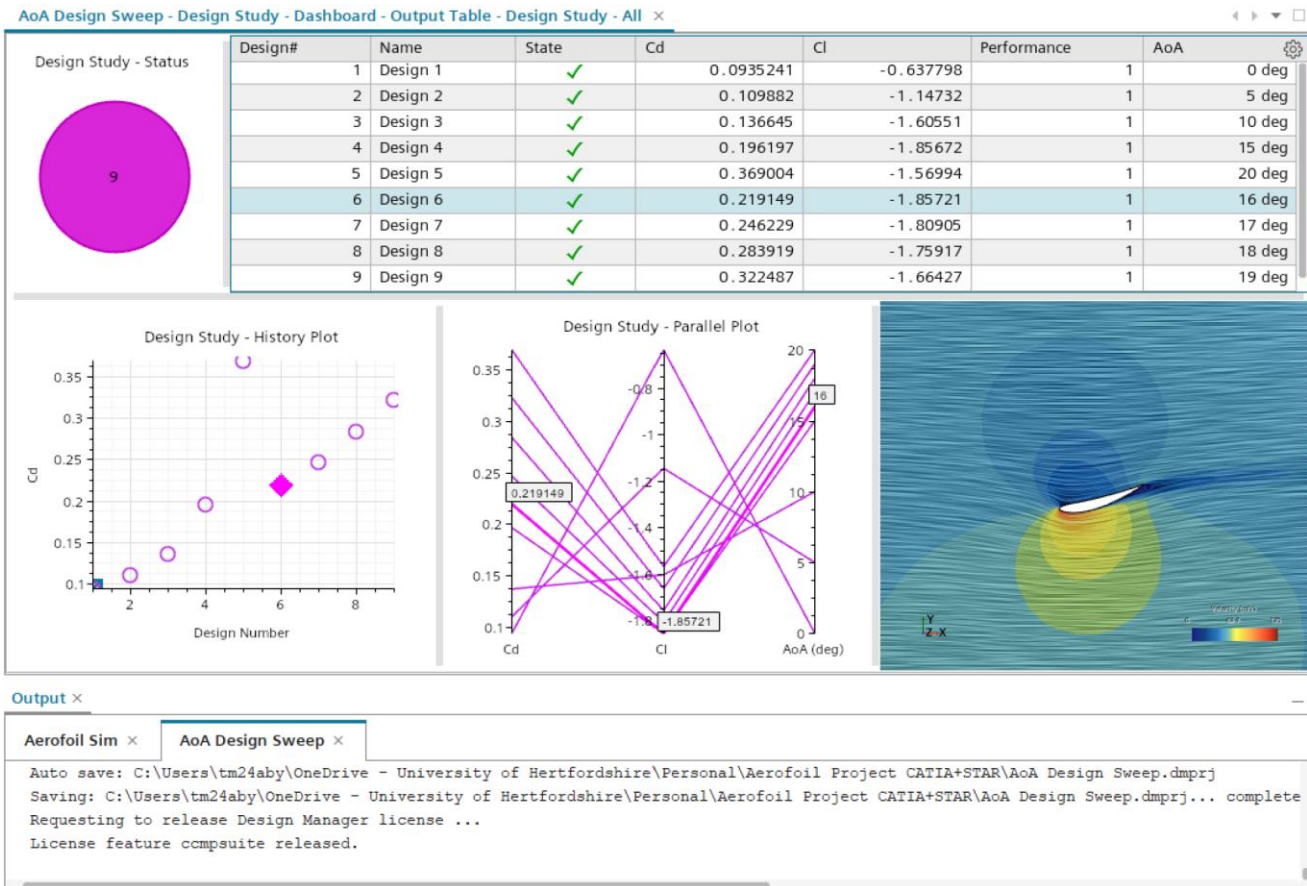


Figure B.1. STAR-CCM+ result dashboard at 16°.

Converting mesh into finite volume representation completed: CPU Time: 0.05, Wall Time: 0.05, Memory: 100.24 MB
 Cells: 6662 Faces: 18597 Vertices: 12210

Mesh Operation Automated Mesh (2D) complete. CPU Time: 0.94, Wall Time: 0.94, Memory: 93.59 MB

Figure B.2. Archived mesh statistics and details.

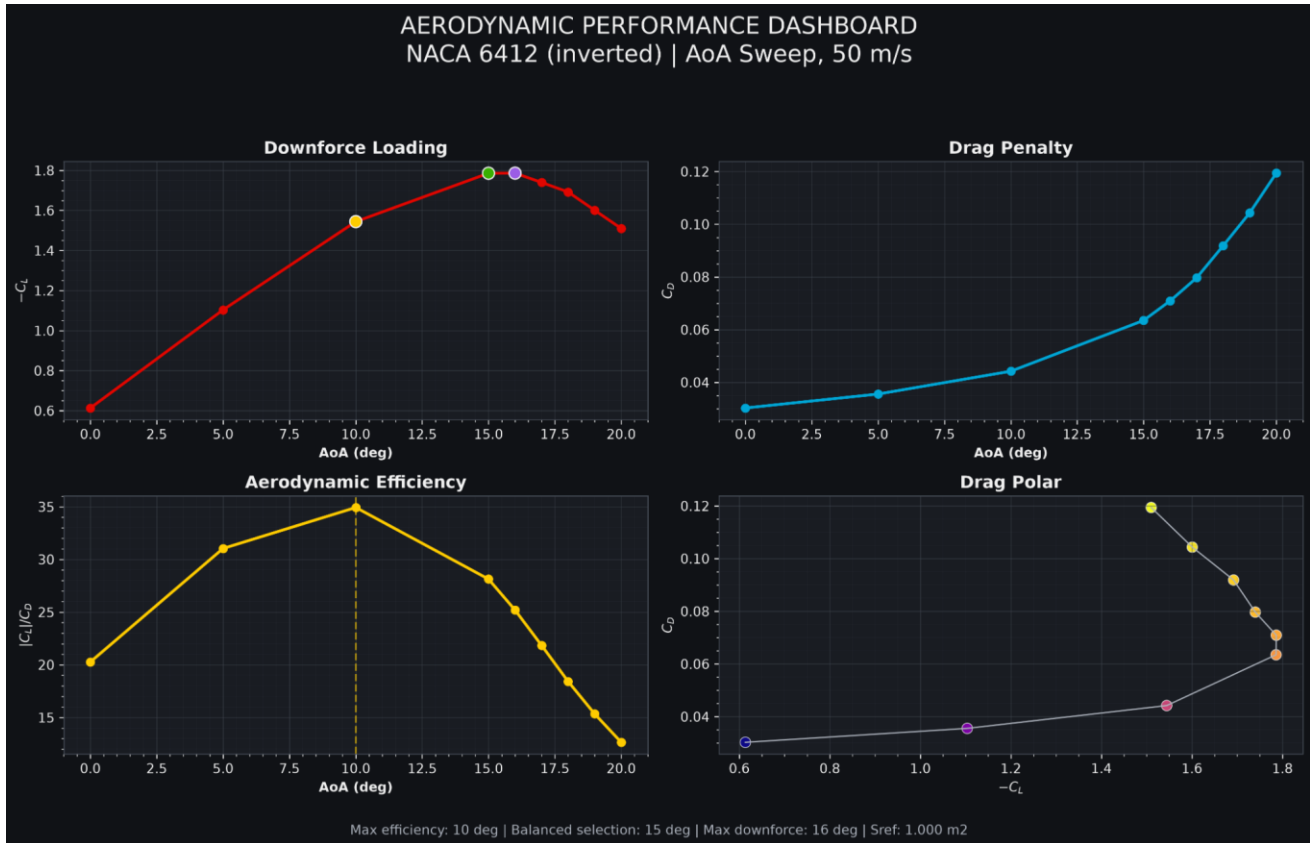


Figure B.3. Full automated performance dashboard.

B.1 Assumption and Evidence Register

ITEM	STATUS
Exact k-ε variant	Unknown from retained evidence
Wall treatment	Unknown from retained evidence
Upper/lower velocity vector	Confirmed as (50, 0, 0) m/s
Force vectors	Downforce (0,1,0); drag (1,0,0)
Iterations	1,000 per design
AoA-sweep mesh	Medium, 6,662 cells
Coefficient basis	Corrected to common 1.000 m ²
XFOIL 15°	Excluded due to non-convergence
Experimental validation	Not performed
Far-field surface control	Confirmed: target and minimum surface sizes of 3.0 m on all four outer boundaries

Appendix C. Project-Plan Compliance Matrix

The matrix confirms how the commitments in the final project plan have been addressed. Where evidence was not retained, the limitation is stated rather than estimated.

PLANNED CONTENT	REPORT LOCATION / STATUS
Introduction and project aim	Sections 1.1 to 1.5
Tool-selection justification	Section 3.1
Excel/CATIA macro automation	Sections 3.2 and 4.2; implemented
STAR-CCM+ automation opportunities	Section 3.4; proposed Java-macro workflow
Python implemented automation	Section 8; implemented and evidenced by the final script
Hand calculations and assumptions	Sections 2.2, 2.3 and 5.1
CAD geometry and export preparation	Section 4; CATPart and STEP workflow described
CFD setup, boundaries and turbulence model	Section 5
Mesh sensitivity and mesh selection	Section 6
Initial and refined AoA studies	Section 7.1
Validation / numerical verification	Section 9
Lift, drag, efficiency and contour discussion	Section 7
Final engineering recommendation	Section 10
Limitations and future improvements	Section 11
Runtime comparison requested in plan	Not available in retained evidence; excluded rather than estimated
Experimental validation	Not completed; declared as future work